

Autonomic Nervous System

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Chemistry),
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Introduction to the Autonomic Nervous system

The peripheral nervous system

- The peripheral nervous system is part of the nervous system which is outside of the central nervous system (CNS: the brain and spinal column).
- There are two subdivisions of the peripheral system:
 - Sensory nerves (nerves which take messages from the body to the CNS)
 - Motor nerves (nerves which carry messages from the CNS to the rest of the body)

Motor nerves of the peripheral nervous system

- These nerves take messages from the CNS to various parts of the body such as **skeletal muscle, smooth muscle, cardiac muscle, and glands.**
- Motor nerves of the peripheral nervous system have been divided into two sub-systems
 - The somatic motor nervous system
 - The autonomic motor nervous system

Somatic motor nervous system

- These are nerves which carry messages from the CNS to skeletal muscles.
- There are no synapses (junctions) and the neurotransmitter at the neuromuscular junction is **acetylcholine**.
- The final result of such messages is **contraction of skeletal muscle**.

The autonomic motor nervous system

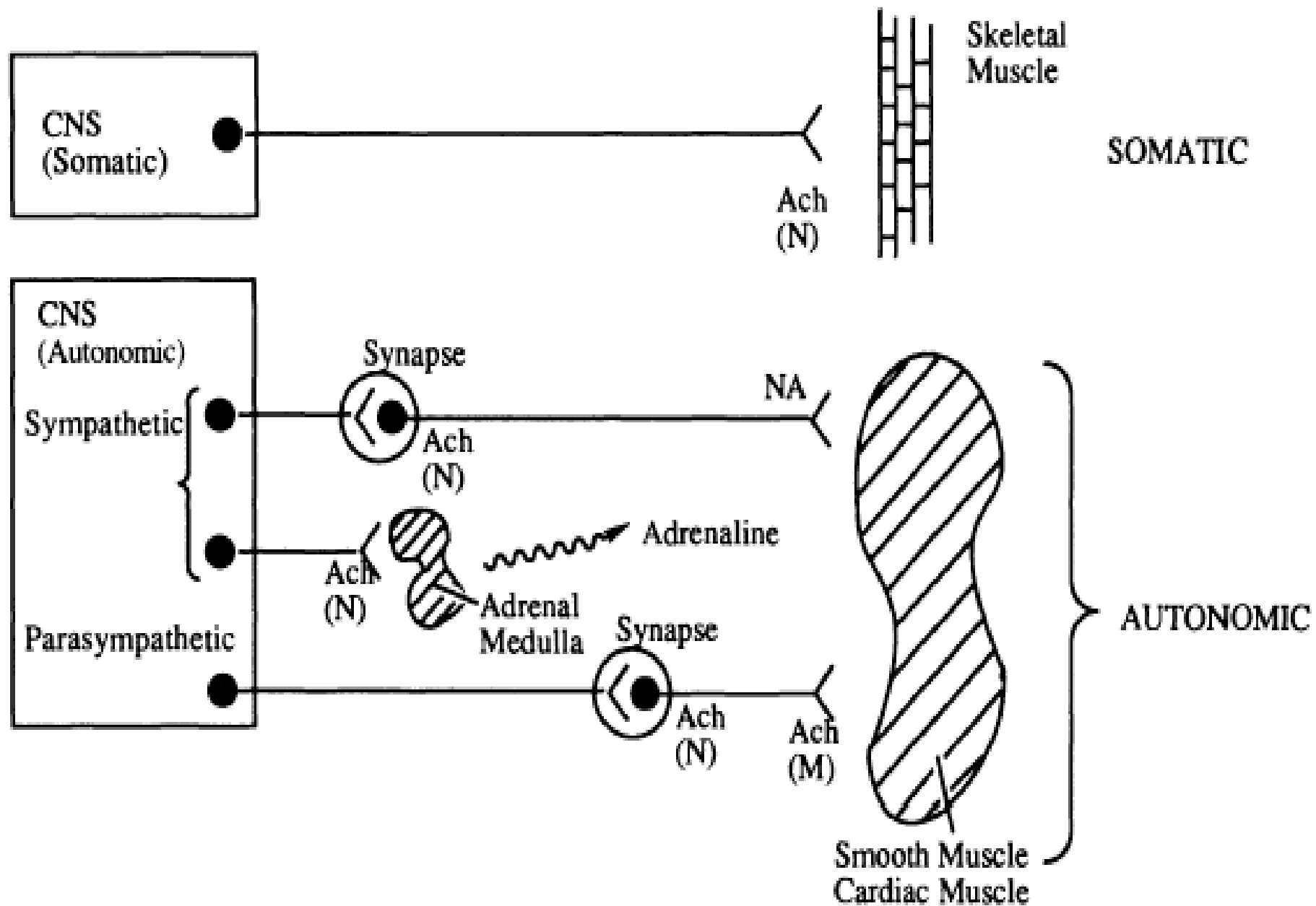
- These nerves carry messages from the CNS to **smooth muscle**, **cardiac muscle**, and the **adrenal medulla**.
- This system can be divided into two subgroups.
 - Sympathetic NS
 - Parasympathetic NS

Parasympathetic nerves

- These leave the CNS, travel some distance, then synapse with a second nerve which then proceeds to the final synapse with smooth muscle.
- The neurotransmitter at both synapses is acetylcholine.

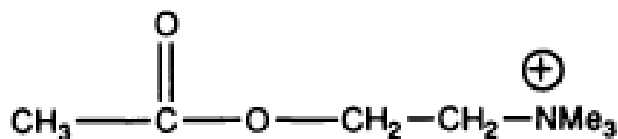
Sympathetic nerves

- These leave the CNS, but almost immediately synapse with a second nerve (neurotransmitter— acetylcholine) which then proceeds to the same target organs as the parasympathetic nerves.
- However, they synapse with different receptors on the target organs and use a different neurotransmitter—noradrenaline

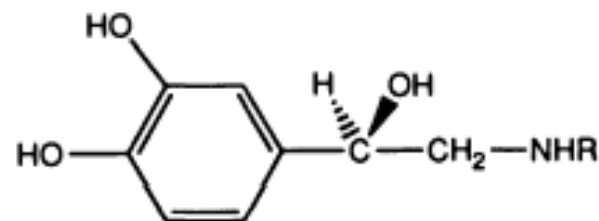


The neurotransmitters

- There are a large variety of neurotransmitters in the CNS, but as far the peripheral nervous system is concerned we need only consider two—**acetylcholine** and **noradrenaline**



Acetylcholine



R = H Noradrenaline
(R = Me Adrenaline)

Actions of the peripheral nervous system

A. Somatic

- Stimulation leads to the contraction of skeletal muscle.

B. Autonomic

- **Sympathetic.**

- Contraction of cardiac muscle and an increase in heart rate.
- Relaxes smooth muscle and reduces the contractions of the GIT and urinary tracts
- Reduces salivation and reduces dilation of the peripheral blood vessels.
- **Fight or flight** response of the body

Parasympathetic.

- The stimulation of the parasympathetic system leads to the opposite effects from those of the sympathetic system.
 - Rest and digest role of the body (house keeping role)
- Acetylcholine is released at the target organs and reacts with receptors specific to it and not to noradrenaline

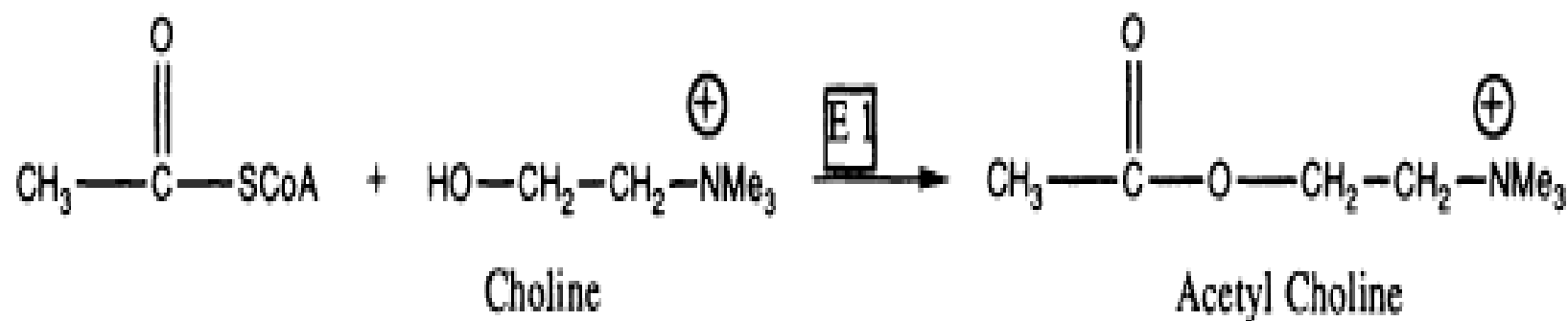
The cholinergic system

- What happens at synapses involving acetylcholine as the neurotransmitter?

Biosynthesis of acetylcholine

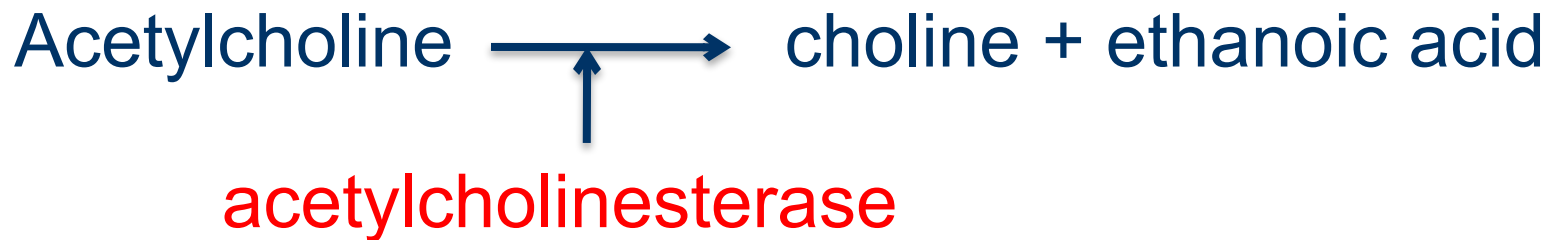
- Acetylcholine is synthesized in the nerve ending of the pre-synaptic nerve from choline and acetyl coA.

- The reaction is catalysed by the enzyme choline-acetyltransferase, E1.

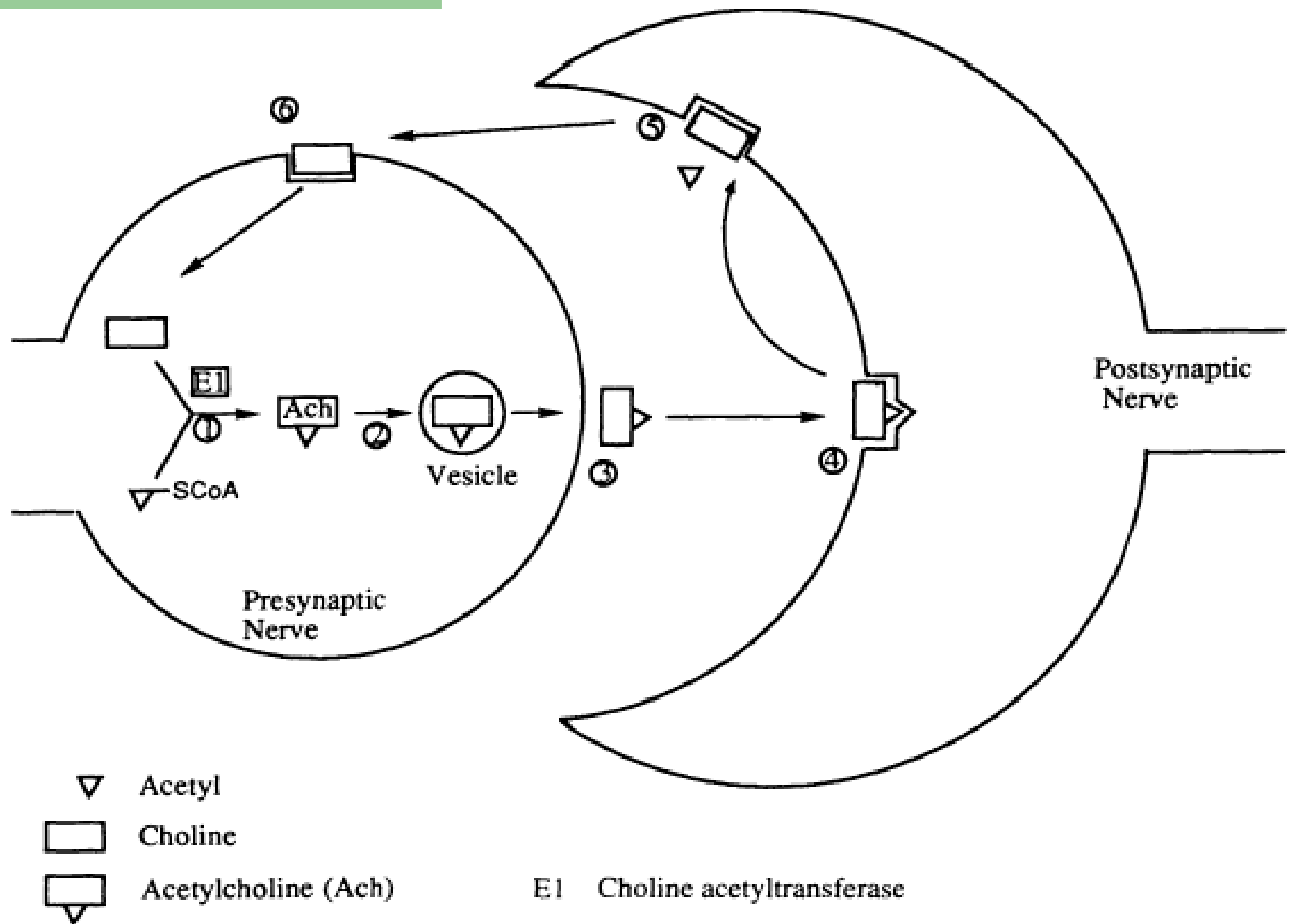


- Acetylcholine is incorporated into membrane-bound vesicles.
- The arrival of a nerve signal leads to the release of acetylcholine.
- Acetylcholine crosses the synaptic gap and binds to the cholinergic receptor leading to stimulation of the second nerve.

- Acetylcholine moves to an enzyme called **acetylcholinesterase** which is situated at the postsynaptic nerve and which catalyses the hydrolysis of acetylcholine to choline and ethanoic acid



- Choline binds to the choline receptor on the presynaptic nerve and is taken up into the cell by an efficient transport system to continue the cycle.



Agonists at the cholinergic receptor

- If there is a lack of acetylcholine acting at a certain part of the body, why do not we just give the patient more acetylcholine?
- After all, it is easy enough to make in the laboratory.

- There are three reasons why this is not feasible.

1. Acetylcholine is easily hydrolysed in the stomach by acid catalysis and cannot be given orally.
2. Acetylcholine is easily hydrolysed in the blood, both chemically and by enzymes (esterases and acetylcholinesterase).
3. There is no selectivity of action. Acetylcholine will switch on all acetylcholine receptors in the body.

- Therefore, we need analogues of acetylcholine which are
 - More stable to hydrolysis
 - More selective with respect to where they act in the body.

There are two ways in which selectivity can be achieved.

- Firstly, some drugs might be distributed more efficiently to one part of the body than another.
- Secondly, cholinergic receptors in various parts of the body might be slightly different.

Acetylcholine

('Fits' both types of receptor)

Ach Analogue

'FIT'

Ach Analogue

Acetylcholine Receptor
Type 1?

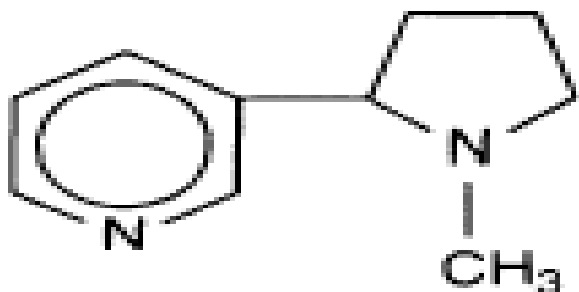
NO 'FIT'

Ach Analogue

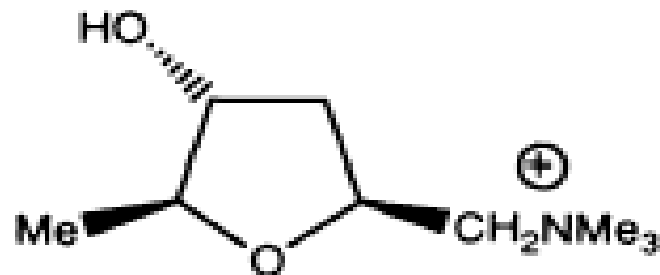
'Wall'

Acetylcholine Receptor
Type 2?

- This is not just a peculiarity of acetylcholine receptors.
 - Subtle differences have been observed for other types of receptors such as those for dopamine, noradrenaline, and serotonin.
- **How do we know if there are different subtypes of acetylcholine receptor?**
 - The first clue came from the action of natural compounds.
- It was discovered that the compounds **nicotine** and **muscarine** were both acetylcholine agonists, but that they had different physiological effects.



NICOTINE



L (+) MUSCARINE

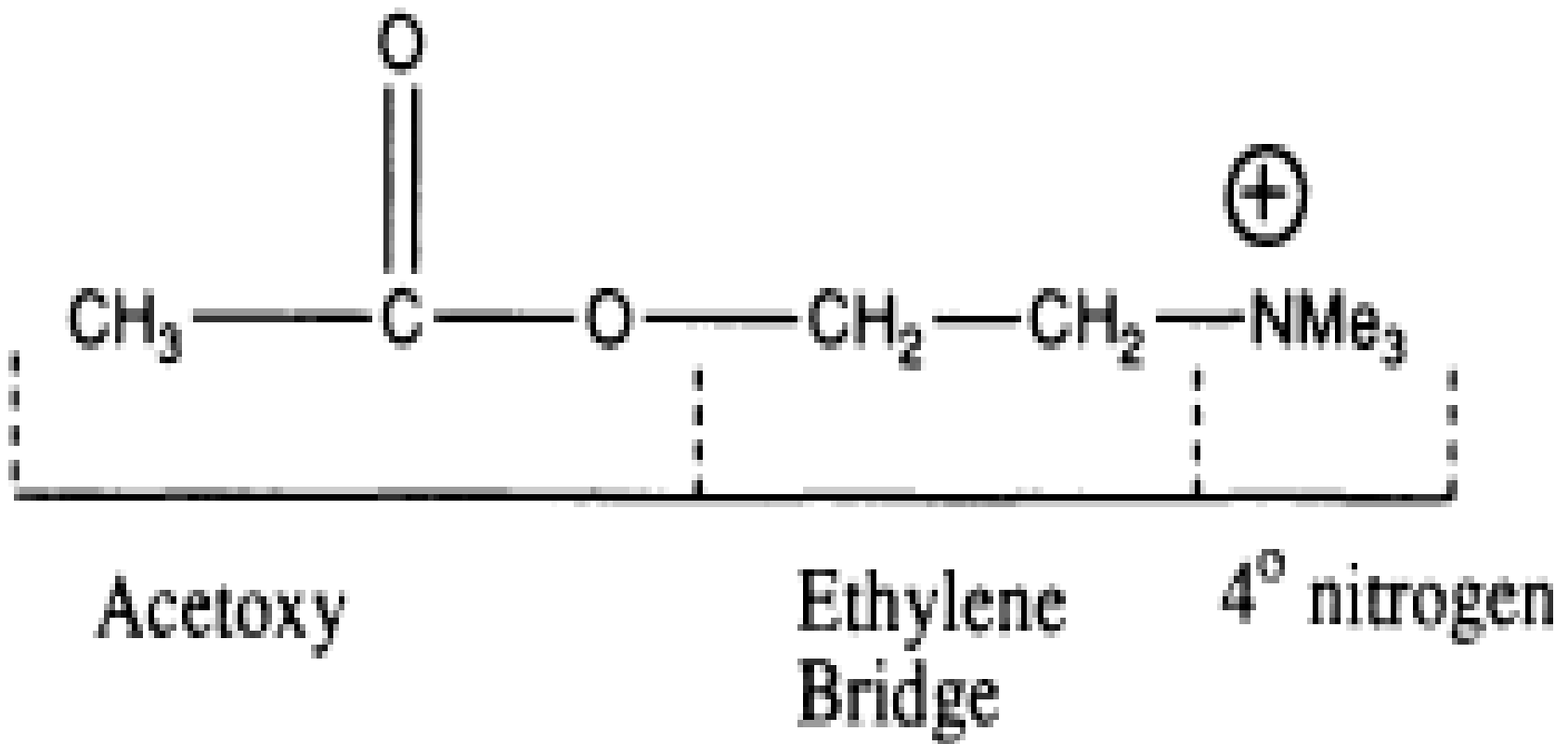
- **Nicotine** was found to be active at the synapses between
 - Two different nerves
 - Nerves and skeletal muscle,
 - But had poor activity elsewhere.

- **Muscarine** was active at the synapses of nerves with
 - Smooth muscle and
 - Cardiac muscle,
 - But showed poor activity at the sites where nicotine was active.

- From these results:
 - There is one type of acetylcholine receptor on skeletal muscles and at nerve synapses
 - Known as **nicotinic receptor, N**
 - There is a different sort of acetylcholine receptor on smooth and cardiac muscles
 - Known as **muscarinic receptor, M**

- Unfortunately, these two compounds are not suitable as medicines since they have **undesirable side-effects**.
- But they have served as a **lead compound** for further development

Acetylcholine: Structure, SAR, and receptor binding



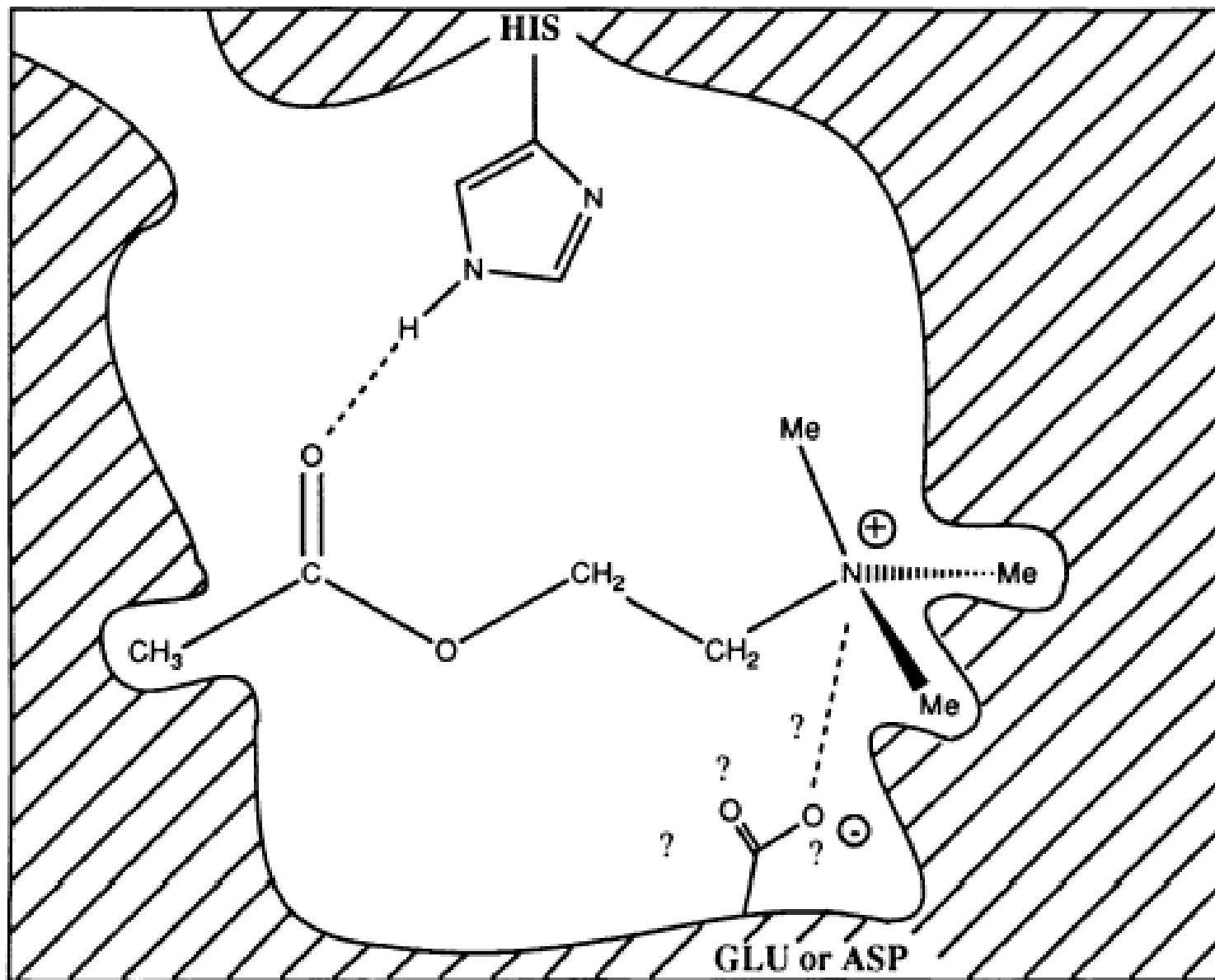
SAR ...

- The positively charged nitrogen atom is essential to activity.
 - Replacing it with a **neutral carbon** atom eliminates activity.
- The distance from the nitrogen to the ester group is important.
- The ester functional group is important.

- The overall size of the molecule cannot be altered greatly.
 - Bigger molecules have poorer activity.
- The ethylene bridge between the ester and the nitrogen atom cannot be extended.
- There must be two methyl groups on the nitrogen.
 - A larger, third alkyl group is tolerated, but more than one large alkyl group leads to loss of activity.

- Bigger ester groups lead to a loss of activity.
- **Conclusions:**
 - Clearly, there is a tight fit between acetylcholine and its binding site which leaves little scope for variation.
 - The above findings fit in with a receptor site as shown in figure below

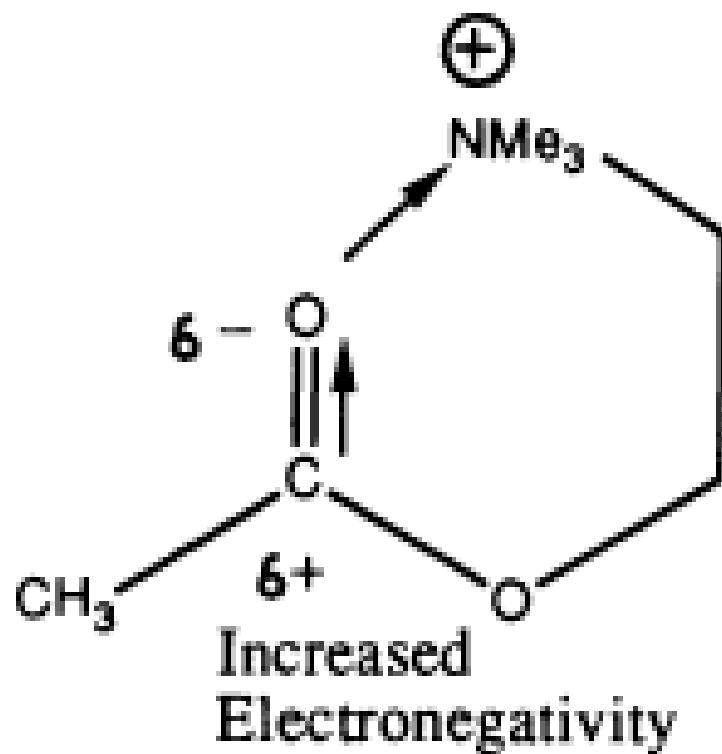
RECEPTOR SITE
(MUSCARINIC)



The instability of acetylcholine

- As described previously, acetylcholine is prone to hydrolysis.
- Why is this and how can the stability being improved?
 - The reason for acetylcholine's instability can be explained by considering one of the conformations that the molecule can adopt

Instability of Ach ...

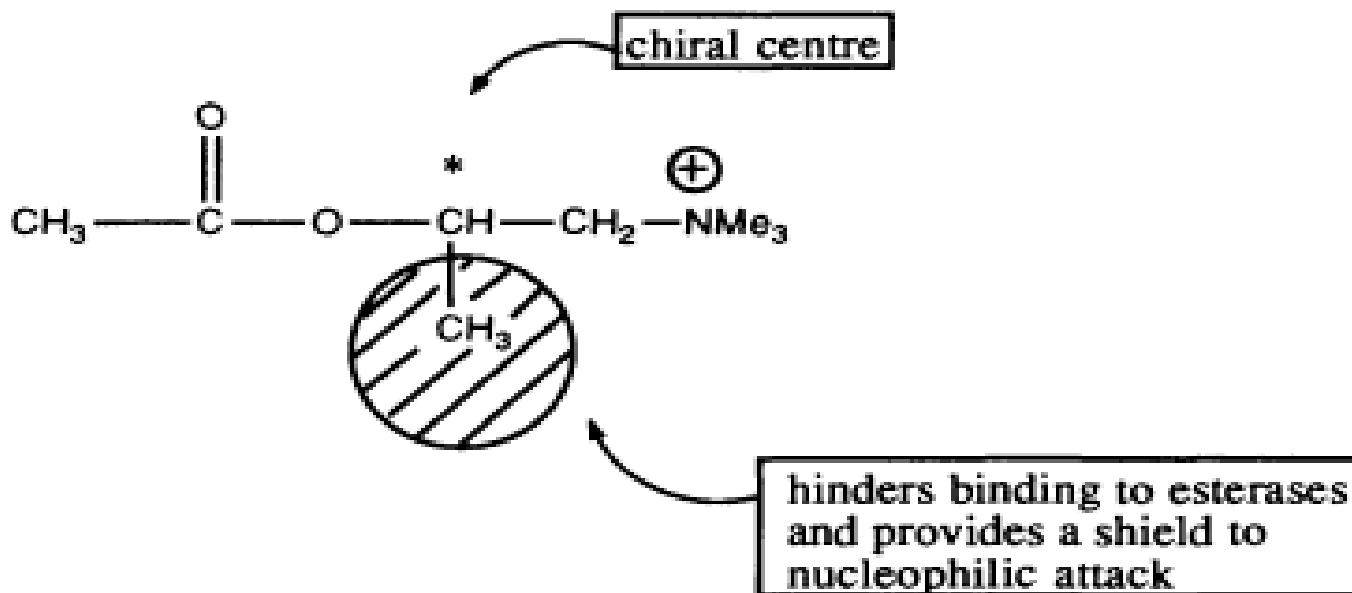


Design of acetylcholine analogues

- In order to tackle the inherent instability of acetylcholine, two approaches are possible:
 - Steric hindrance
 - Electronic stabilization

I. Steric hindrance

- The principle involved here can be demonstrated with methacholine



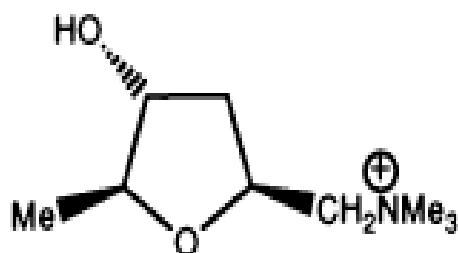
- This analogue of acetylcholine contains an extra methyl group on the ethylene bridge.
- The reasons for putting it there are twofold.
 - Firstly, it is to try and build in a shield for the carbonyl group.
 - The bulky methyl group should hinder the approach of any potential nucleophile and slow down the rate of hydrolysis.
 - It should also hinder binding to the esterase enzymes, thus slowing down enzymatic hydrolysis.

- The results were encouraging, with methacholine proving **three times more stable** to hydrolysis than acetylcholine.
- Why don't we put on a bigger alkyl group? Alternatively, why not put a bulky group on the acyl half of the molecule?

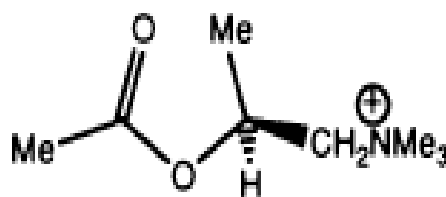
- In fact, these approaches were tried, but **failed**
- The fit between acetylcholine and its receptor is so tight that there is little scope for enlarging the molecule.
- The extra methyl group is as much as we can get away with.
- Larger substituents certainly cut down the chemical and enzymatic hydrolysis, but they also prevent the molecule binding to the cholinergic receptor.

- **In conclusion**, attempts to increase the steric shield beyond the methyl group certainly increase the stability of the molecule, but decrease its activity since it **cannot** fit the cholinergic receptor.
- One other very useful result was obtained from methacholine.
 - It was discovered that the introduction of the methyl group led to **significant muscarinic activity** and **very little nicotinic activity**.

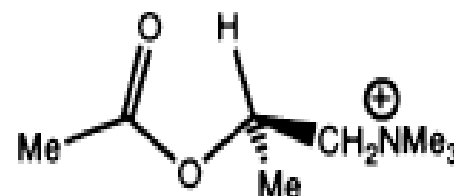
- This result is perhaps more important than the gain in stability.
 - The good binding to the muscarinic receptor can be explained if we compare the active conformation of methacholine with muscarine.



MUSCARINE



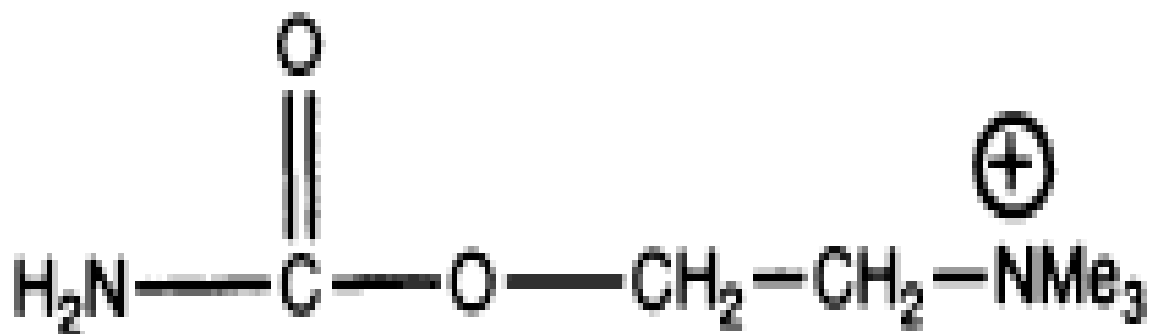
(S)



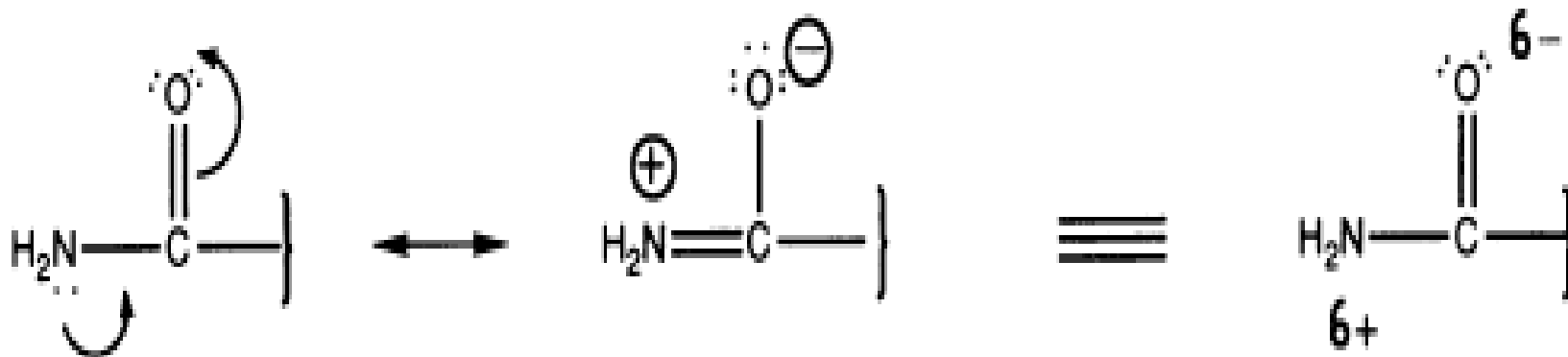
(R)

Electronic effects

- The best example of this approach is provided by **carbachol**, a long acting cholinergic agent which is resistant to hydrolysis.
- In carbachol, the acyl **methyl** group has been replaced by an **NH₂** group which is of comparable size and can therefore fit the receptor



- The resistance to hydrolysis is due to the electronic effect of the carbamate group.



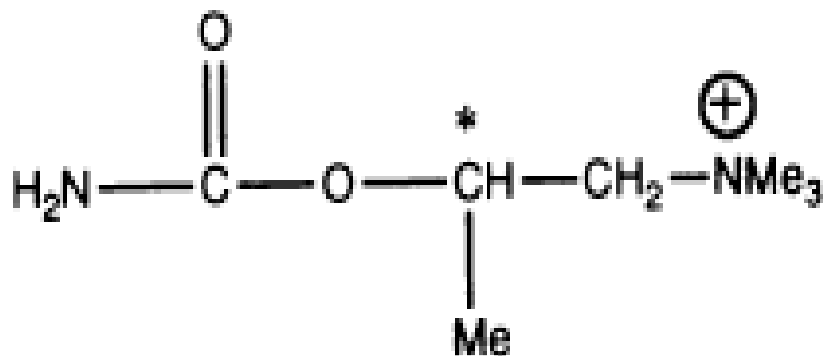
- Carbachol is certainly stable to hydrolysis and is the right size to fit the cholinergic receptor, but it is by no means a foregone conclusion that it will be active.
- After all, a hydrophobic methyl group has been replaced with a polar NH_2 group and this implies that a polar group has to fit into a hydrophobic pocket in the receptor.

- Fortunately, carbachol does fit and is active.
- Since the methyl group of acetylcholine has been replaced with an amino group without affecting the biological activity, we can call the amino group a '**bioisostere**' of the methyl group.
 - Therefore, the inclusion of an electron donating group such as the amino group has greatly increased the chemical and enzymatic stability of our cholinergic agonist.

- It is found that carbachol shows **very little selectivity** between the muscarinic and nicotinic sites.
- Carbachol is used clinically for the treatment of glaucoma.
 - The drug is applied locally and so selectivity is not a great problem.

Combining steric and electronic effects

- The compound obtained by combining the above two effects is **bethanechol** which is both stable to hydrolysis and selective in its action.



Bethanechol

- **Bethanechol** is used therapeutically in stimulating the gastrointestinal tract and urinary bladder after surgery.
 - Both these organs are '**shut down**' with drugs during surgery

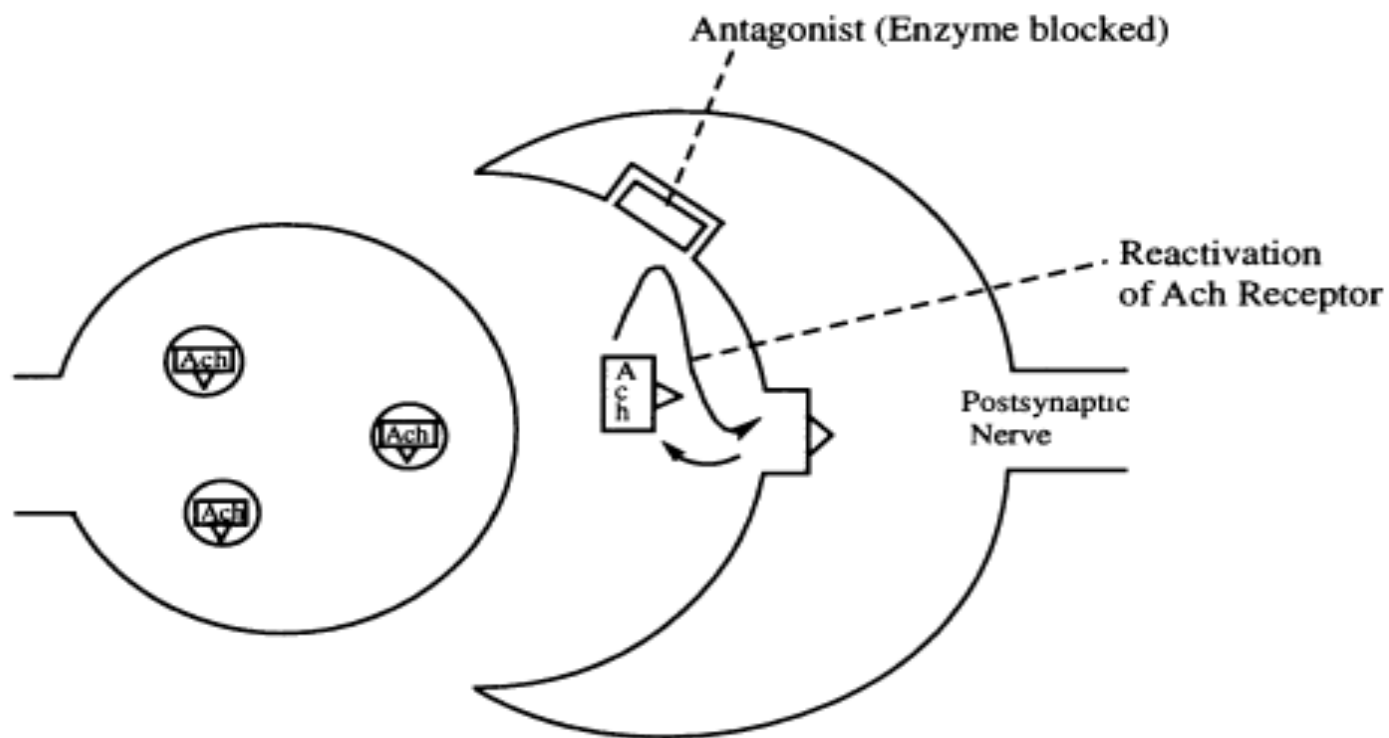
Acetylcholinesterase inhibitors (indirectly acting cholinergic agonists)



Anticholinesterase and Acetylcholinesterase

- **Acetylcholinesterase** is the enzyme which hydrolyses acetylcholine.
- Anticholinesterases are antagonists of this enzyme
- Antagonist at the acetylcholinesterase enzyme will have the same biological effect as an agonist at the cholinergic receptor.

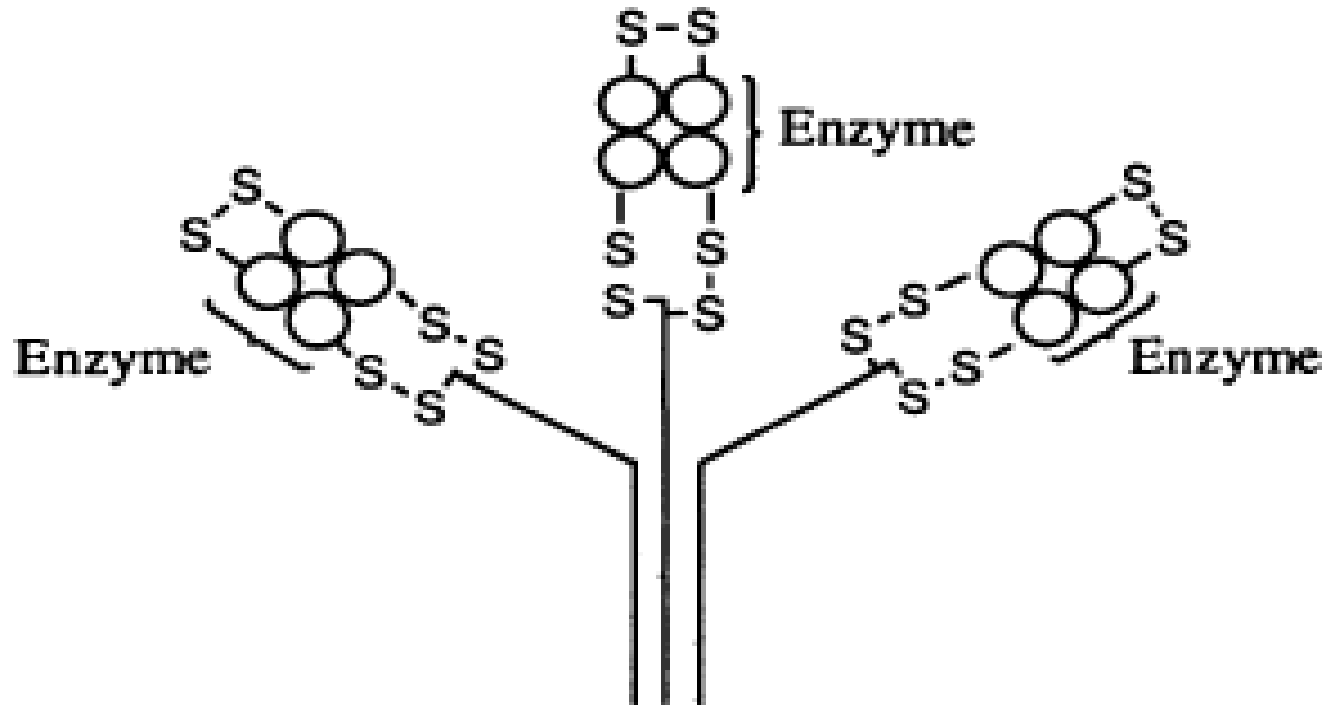
Anticholinesterases...



Structure of the acetylcholinesterase enzyme

- Has a fascinating tree-like structure
 - The trunk of the tree is a collagen molecule
 - There are three branches (disulfide bridges) leading off from the trunk
 - The enzyme itself is made up of four protein subunits, each of which has an active site

Structure of the...



Design of anticholinesterases

- The design of anticholinesterases depends on
 - The shape of the enzyme active site,
 - The binding interactions involved with acetylcholine, and
 - The mechanism of hydrolysis.

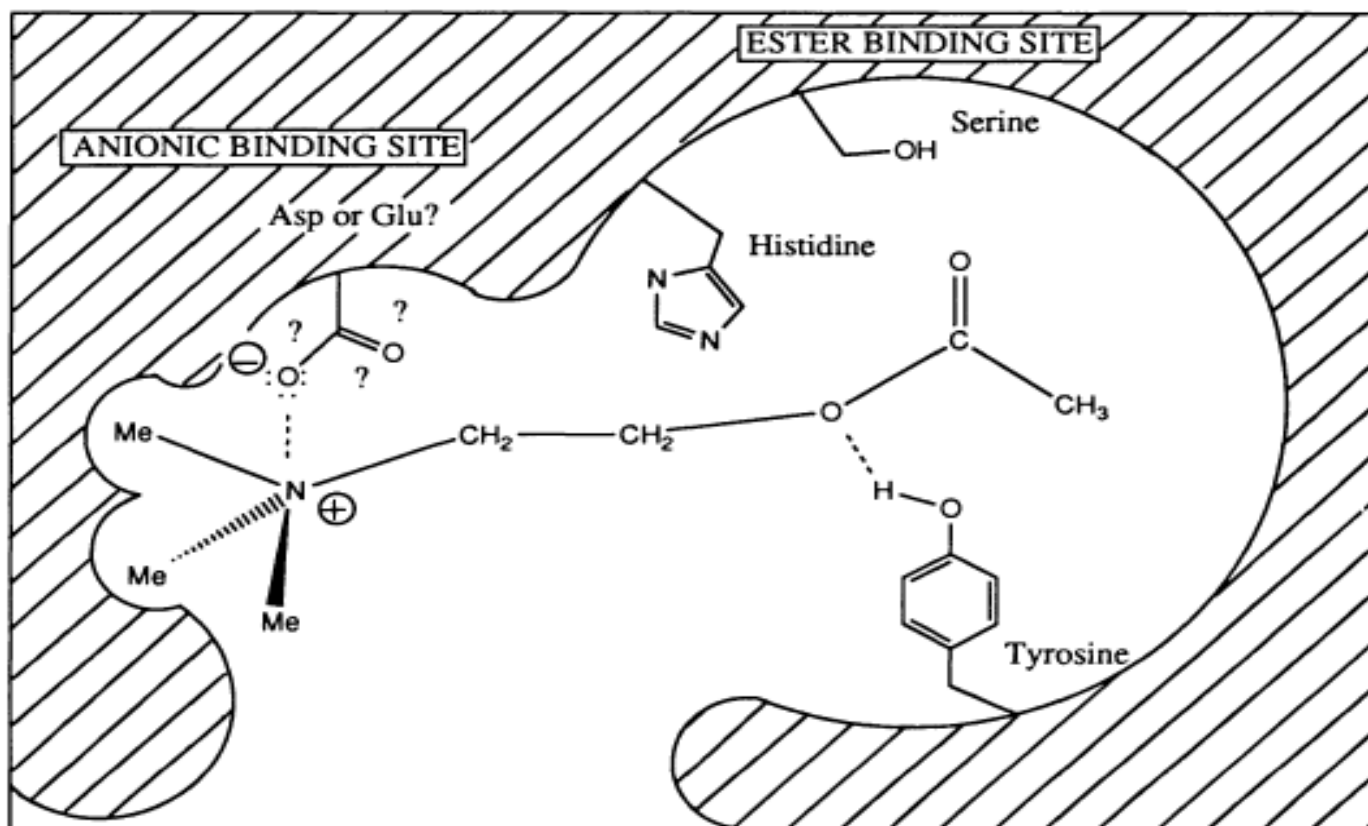
Binding interactions at the active site

- There are two important areas to be considered
 - The ***anionic*** binding site and the ***ester*** binding site
- Note that:

Acetylcholine binds to the cholinesterase enzyme by

 - (a) Ionic bonding to an Asp or Glu residue,
 - (b) Hydrogen bonding to a tyrosine residue

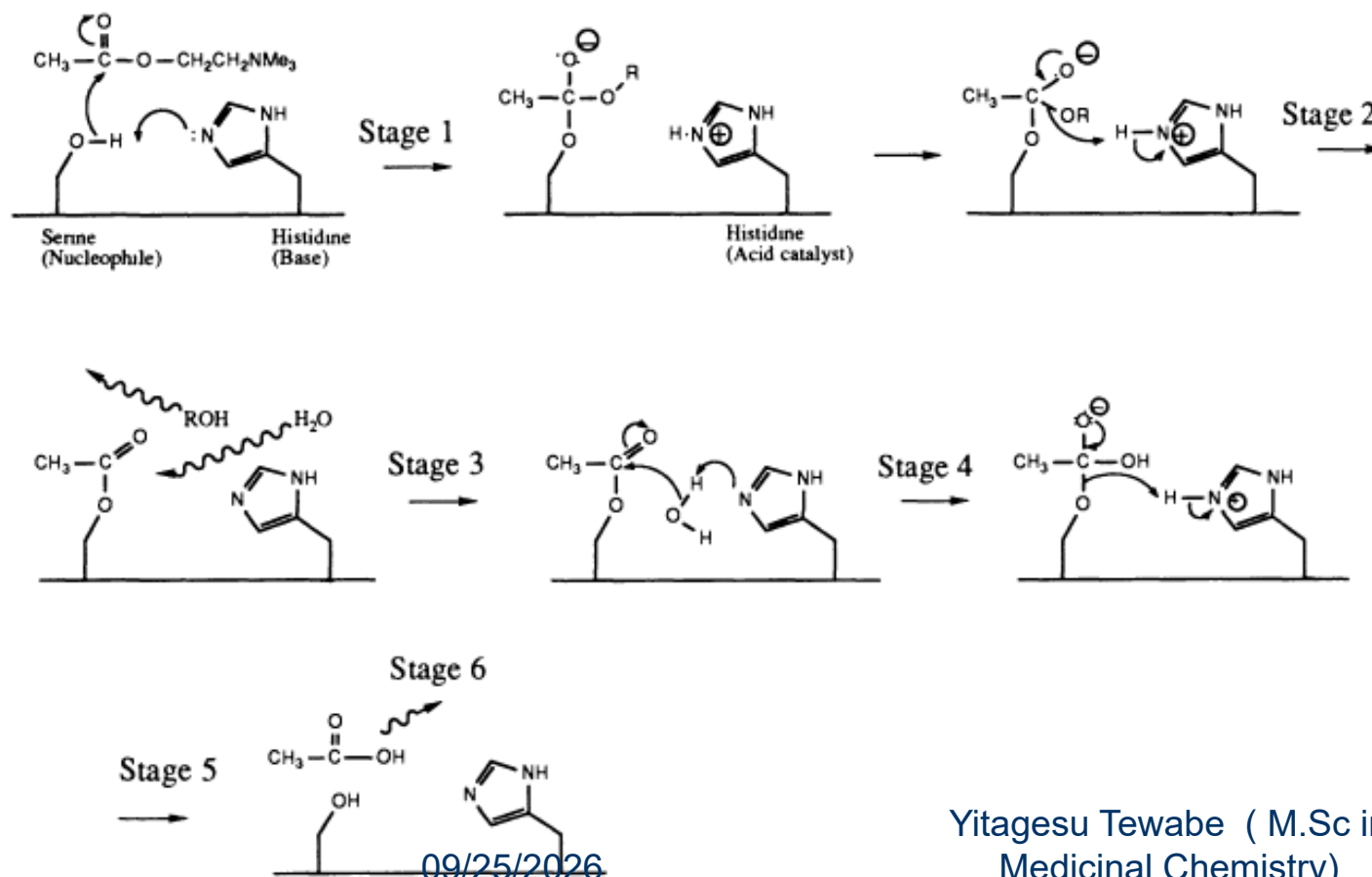
Binding interactions...



Mechanism of hydrolysis

- The histidine residue acts as an acid/base catalyst throughout the mechanism
- Serine plays the part of a nucleophile (but is poor nucleophile)

Stages to the mechanism



Stages to ...

- Enzymatic hydrolysis by cholinesterase is one hundred million times faster than chemical hydrolysis.
- The process is so efficient that acetylcholine is hydrolysed within a hundred microseconds of reaching the enzyme

Anticholinesterase drugs

- Anticholinesterase drugs stop the enzyme from hydrolysing acetylcholine
- This antagonism can be
 - Reversible
 - Irreversible
- There are two main groups of acetylcholinesterase inhibitors
 - i. Carbamates
 - ii. Organophosphorus agents

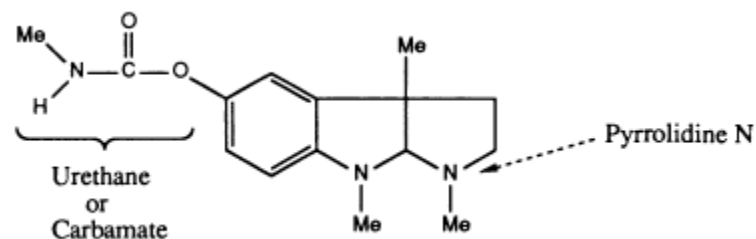
i. The carbamates

Physostigmine

- Is a natural product
- Was discovered in 1864 as a product of the poisonous calabar beans from West Africa
- The structure was established in 1925

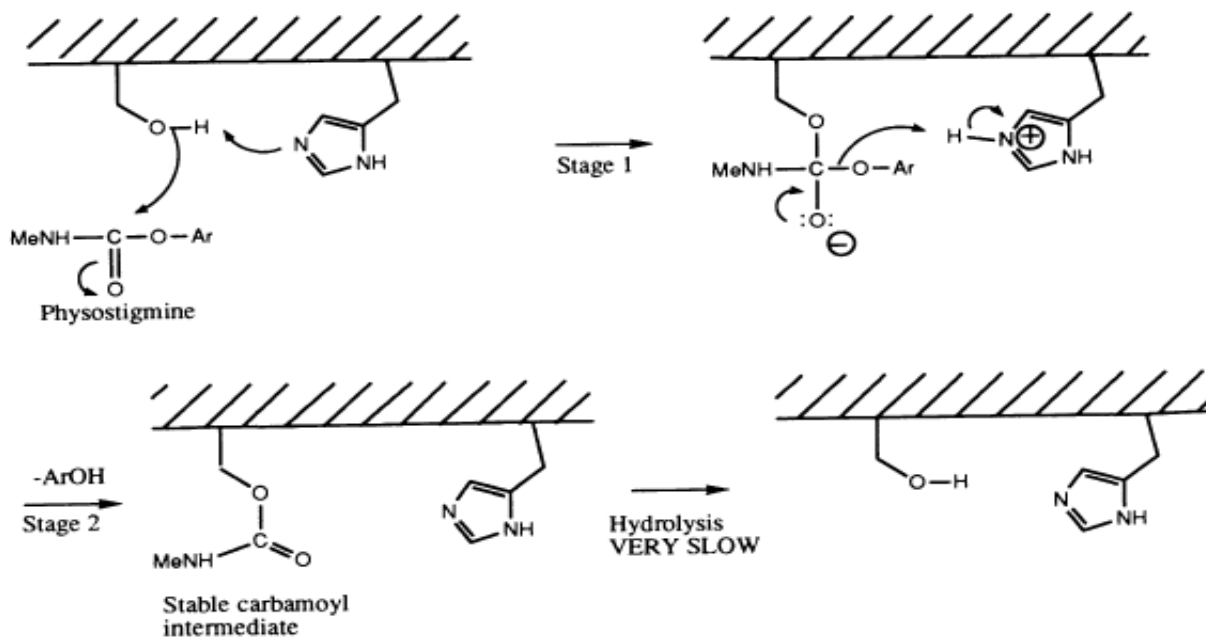
The carbamates...

Structure-activity relationships (SAR)



- The carbamate group is essential to activity
- The benzene ring is important
- The pyrrolidine nitrogen (which is ionized at blood pH) is important

Physostigmine's antagonistic properties

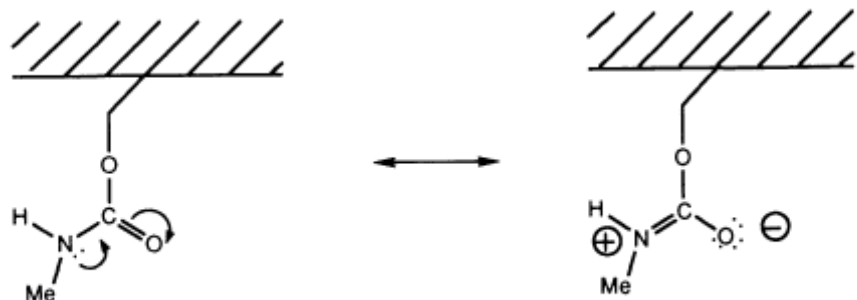


Physostigmine's antagonistic...

- The first three stages proceed as normal with the above steps that we have seen for acetylcholine. However; the next stage turns out to be extremely slow.

Why is this final stage so slow?

- The carbamoyl/enzyme intermediate is stabilized because the nitrogen can feed a lone pair of electrons into the carbonyl group



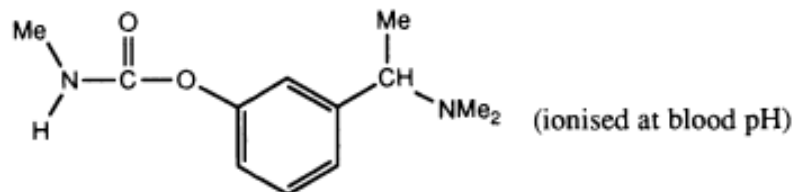
Carbonyl group
'deactivated'

Yitagesu Tewabe (M.Sc in
Medicinal Chemistry)

Analogues of physostigmine

- Physostigmine has limited medicinal use
 - only been used in the treatment of glaucoma
- Simpler analogues have been made and have been used in the treatment of **myasthenia gravis** and as an **antidote to curare**.
- They include

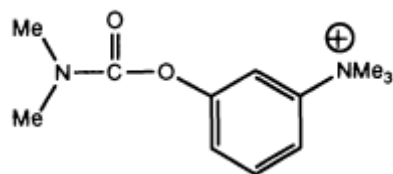
Miotine



Analogues of...

- Miotine suffers from the following disadvantages:
 - susceptible to chemical hydrolysis
 - CNS side effect

Neostigmine

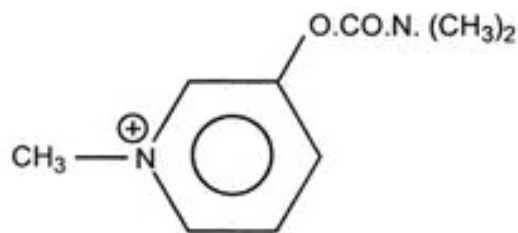


- ✓ Was designed to deal with both the problems described above.

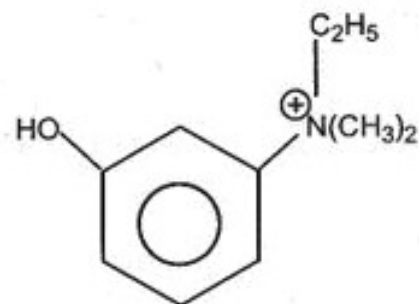
Analogues of...

- Two further points to note about neostigmine
 - The quaternary nitrogen is 4.7 Å away from the ester
 - The direct bonding of the quaternary centre to the aromatic ring reduces the number of conformations which the molecule can take up.
- Neostigmine has proved a useful agent and is still in use today

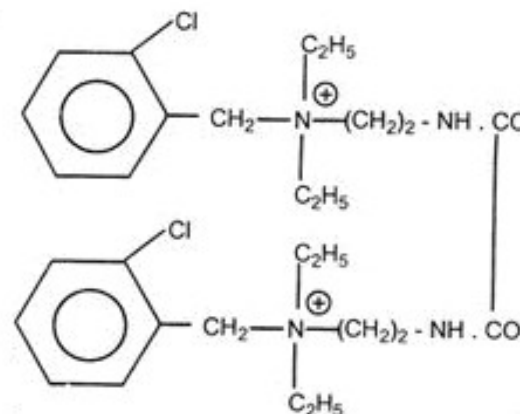
Other Reversible Cholinesterase Inhibitors



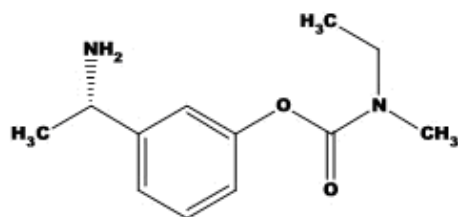
Pyridostigmine



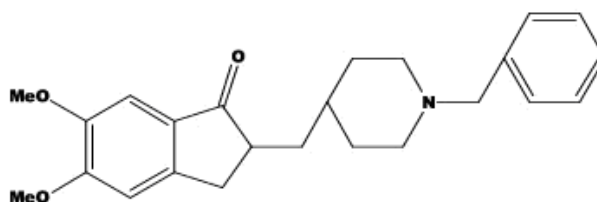
Edrophonium



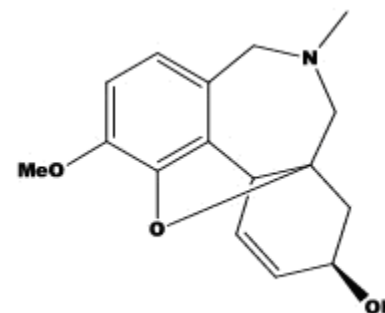
Ambenonium



Rivastigmine



Donepezil

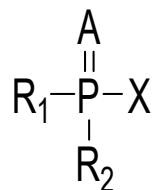


Galantamine

Irreversible Acetylcholinesterase Inhibitors

- AChE is inhibited irreversibly by a group of phosphate esters that are highly toxic.
- These chemicals are nerve poisons and have been used in warfare, in bioterrorism, and as agricultural insecticides.

- A general formula of such compounds:



where R_1 = alkoxy

R_2 = alkoxy, alkyl, or tertiary amine

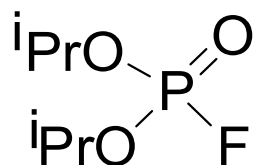
X = good leaving group (e.g., F, CN, thiomalate, *p*-nitrophenyl)

When A is sulfur, bioactivation is required before the compound becomes effective as an inhibitor of cholinesterase.

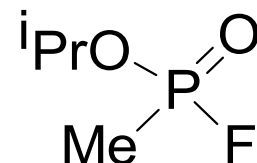
- Inhibition of AChE by organophosphorus compounds takes place in two steps
 - Association of enzyme and inhibitor and Phosphorylation step, completely analogous to acylation by the substrate.

1. Nerve gas

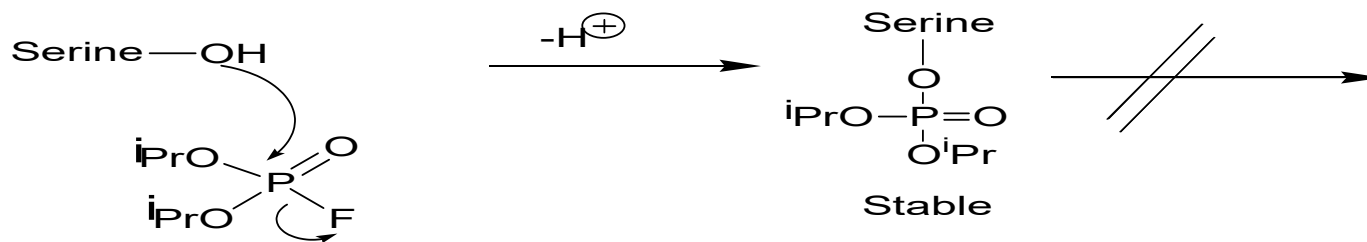
- The nerve gases dyflos and sarin were discovered and perfected long before their mode of action was known.
- Dyflos was developed as a nerve gas in the Second World War.
 - It inhibits AChE by irreversibly phosphorylating the serine residue at the active site.



DYFLOS (Diisopropyl fluorophosphate)



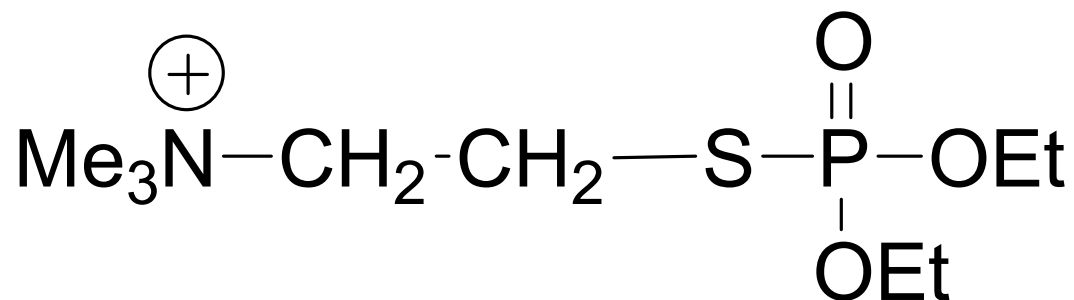
SARIN



- Phosphorylated enzyme is extremely resistant to hydrolysis

2. Medicine

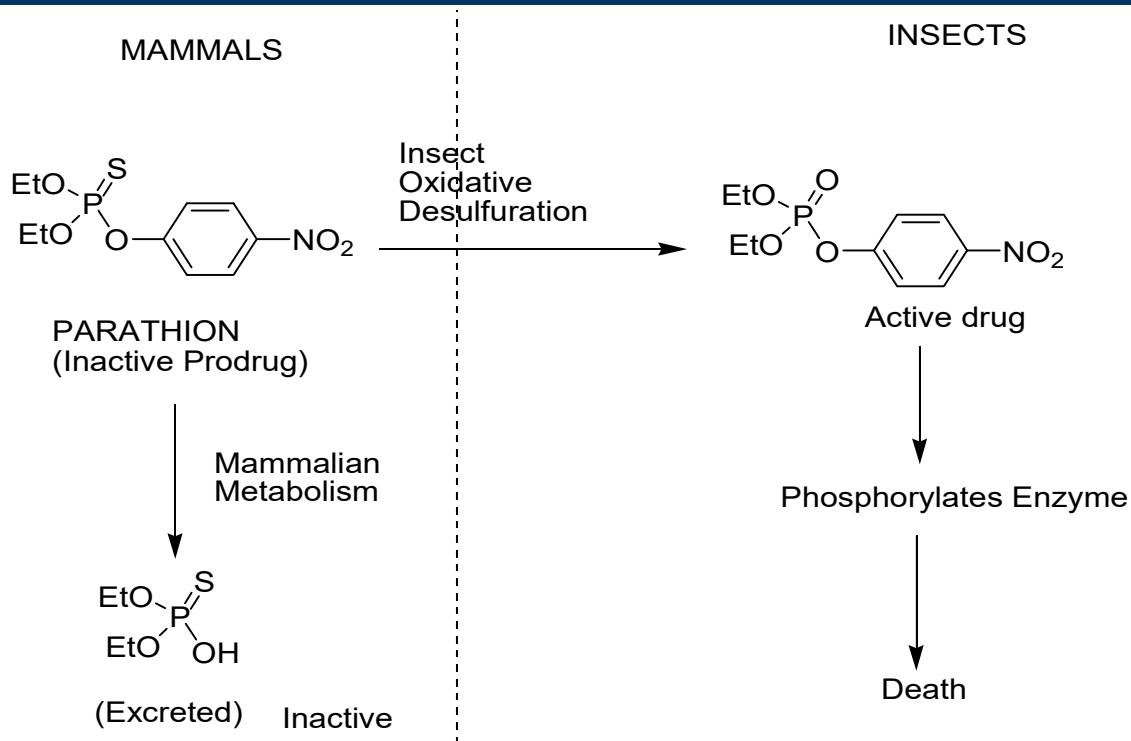
- Echothiophate were designed to fit the active site more accurately by including a quaternary amine to bind with the anionic region.



- This means that lower doses would be more effective.
- Echothiopate is used medicinally in the form of eye drops for the treatment of glaucoma and has advantages over dyflos.
- Unlike dyflos, echothiopate slowly hydrolyses from the enzyme over a matter of days.

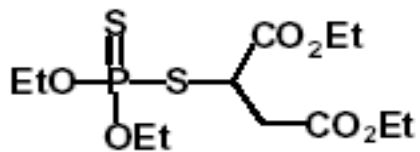
3. Insecticides

- Parathion and malathion are relatively non-toxic with respect to nerve gases.
- The phosphorus-sulfur double bond prevents these molecules from antagonizing the active site on the acetylcholinesterase enzyme.
- The equivalent compounds containing a phosphorus-oxygen double bond are, on the other hand, lethal compounds.



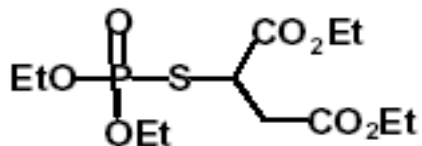
Metabolization of insecticides in mammals and insects

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Medicinal Chemistry)

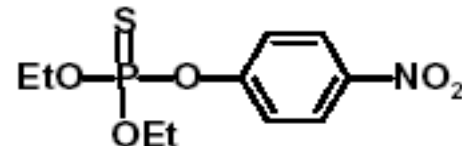


Malathion
(Weak AChE inhibitor)

↓
Microsomal
oxidation

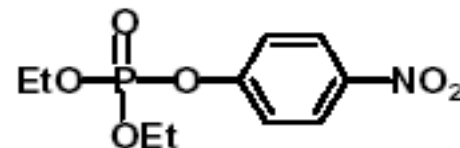


Malaoxon
(10,000 times more active
AChE inhibitor)



Parathion
(Weak AChE inhibitor)

↓
Microsomal
oxidation



paraoxone
(High AChE inhibitory
activity)

Mechanism of Action

- The chemical logic involved in the development of effective AChEIs was to synthesize compounds that would be substrates for AChE and
 - Result in an acylated enzyme more stable to hydrolysis than a carboxylate ester.
- Phosphate esters are very stable to hydrolysis, being even more stable than many amides.

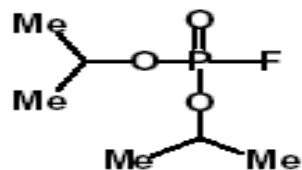
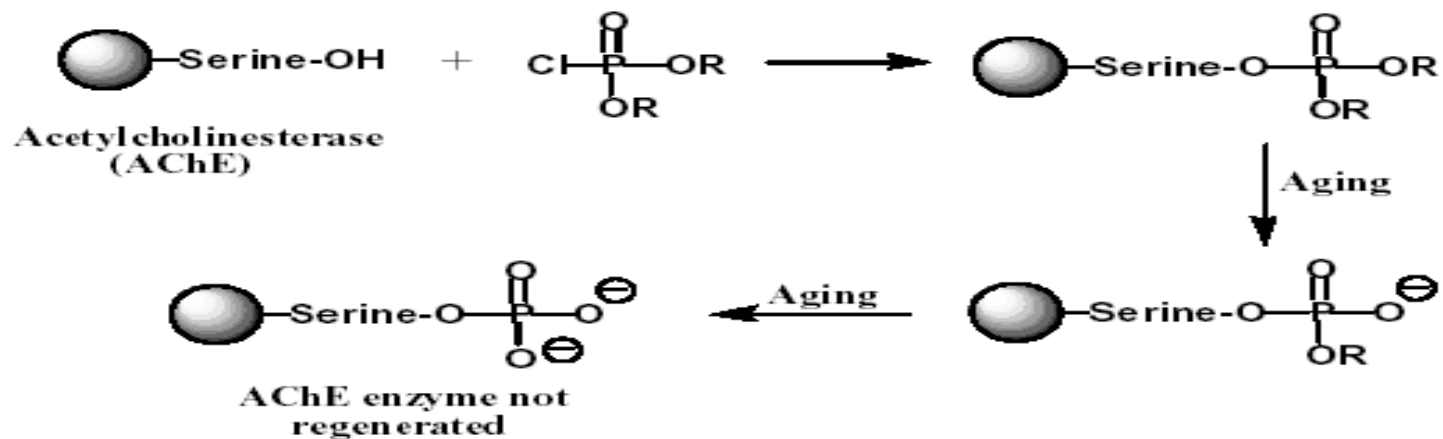
- Application of this chemical property to the design of AChEI compounds led to derivatives of phosphoric, pyrophosphoric and phosphonic acids that are effective inhibitors of AChE.
- These act as inhibitors by the same mechanism as the carbamate inhibitors except that they leave the enzyme esterified as the phosphate esters.

- The rate of hydrolytic regeneration of the phosphorylated enzyme is much slower than that of the carbamylated enzyme and its rate is measured in hours.
- Because duration of action of these compounds is much longer than that of the carbamate esters, they are referred to as irreversible inhibitors of AChE.

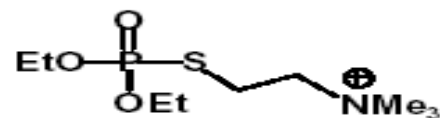
- An important difference between irreversibly phosphoester derived AChEs and reversible AChEs is that the phosphorylated AChE can undergo a process known as *aging*.
 - Aging is the result of cleavage of one or more of the phosphoester bonds while the AChE is phosphorylated.

- This reaction affords an anionic phosphate that possesses a phosphorus atom which is much less electrophilic and therefore, much less likely to undergo hydrolytic regeneration than the original phosphoester.
- Thus, the aged phosphorylated enzyme does not undergo nucleophilic attack and regeneration by antidote for phosphate ester AChEIs.

- This aging process occurs over a period of time, which depends on the rate of the P-O bond cleavage reaction; during this time, the antidotes to phosphate ester poisoning may be effective.



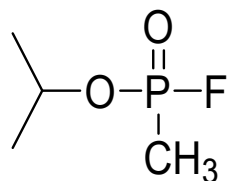
Diisopropylfluorophosphate



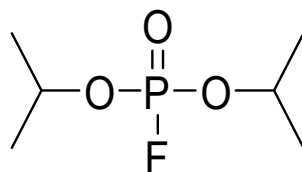
Ecothiophosphate

- Only those phosphorus-derived AChEIs that possess at least one phosphoester group undergo this aging process.
- Knowledge of the chemical mechanisms associated with irreversible inhibition of AChE and the aging process led to the development of deadly phosphorus-derived chemical warfare agents, one of which is sarin.

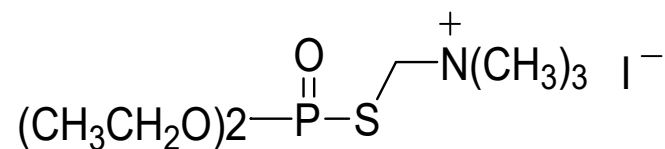
- When this compound phosphorylates AChE, only one aging reaction takes place, and then the enzyme becomes completely refractory to regeneration by the currently available antidotal agents.



Sarin



Diisopropylfluorophosphate
(DFP)



Echothiophate Iodide

Cholinesterase Activators (Antidote for Irreversible AChEIs)

- Water is a nucleophile capable of hydrolyzing acetylated AChE rapidly and regenerating the active enzyme.
- Phosphorylated AChE (irreversibly inhibited), however, was known to involve a phosphate ester of serine.
- The rate of hydrolysis is much slower for organic phosphate esters than carboxylate esters, and significantly stronger nucleophile than water would be required for efficient hydrolysis of phosphate esters.

- **Problem:** design of reagents capable of efficiently catalyzing phosphate ester hydrolysis and regenerating active AChE, while being safe enough for use as therapeutic agents.
- **Hydroxylamine** ($\text{NH}_2\text{-OH}$) is a strong nucleophilic compound that efficiently cleaves phosphate esters and increases significantly the rate of hydrolysis of phosphorylated AChE but only at toxic concentration.

Development of structural analog of hydroxylamine without toxicity

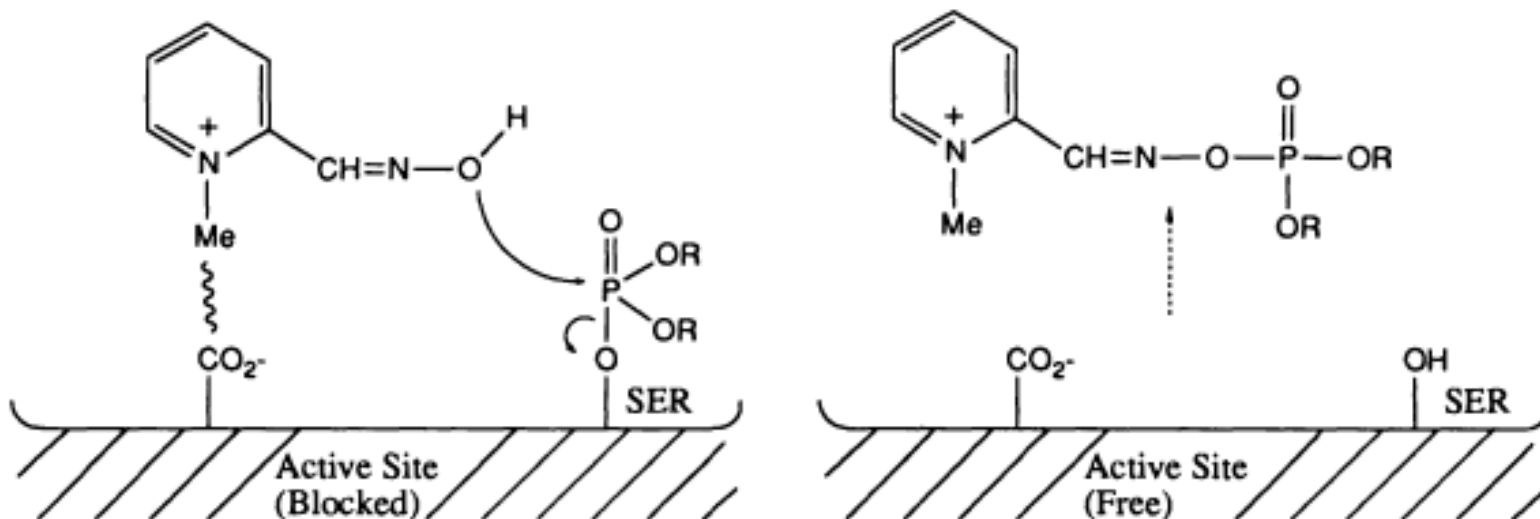
- Logical approach is to design compound with
 - (1) a high degree of selectivity and strong binding affinity for AChE and
 - (2) carry a hydroxylamine-like nucleophile into close proximity to the phosphorylated serine residue.
- **Approach:** synthesis of hydroxylamine derivatives of organic compounds possessing a functional group bearing a positive charge.

- The reaction of hydroxylamine with aldehydes or ketones affords oximes, which possess the desired nucleophilic oxygen atom.
 - *Pyridine ring* is an attractive carrier for the oxime function because such groups are common in a number of biochemical systems (e.g., NAD, NADP), indicating a possible lower order of toxicity.

- There are three readily available positional isomers of pyridine aldehyde that can be converted easily to oximes.
 - Of the three isomeric pyridine aldoxime methiodides, the most effective is the isomer derived from 2-pyridinylaldehyde (2-pyridine aldoxime methyl chloride (Pralidoxime, 2-PAM)).
- Finally, the nitrogen atom of the pyridine ring can be converted to a quaternary ammonium salt by treatment with methyl iodide.

- This cationic charge increases affinity of the compound for the “anionic binding site” of the phosphorylated AChE.
- Proposed mechanism for regeneration of AChE
 - Binding of the quaternary ammonium nitrogen of 2-PAM to the “anionic binding site” of phosphorylated AChE, which places nucleophilic oxygen of 2-PAM in close proximity to the electrophilic phosphorus atom

- Nucleophilic attack of the oxime oxygen results in breaking of the ester bond between the serine oxygen atom and the phosphorus atom, resulting in regenerated active form of AChE and phosphorylated 2-PAM



- Pralidoxime must be given within a short period of time, after enzyme phosphorylation generally a few hours, for it to be effective because of the aging process of the phosphorylated enzyme.
- After aging has occurred, 2-PAM will not regenerate the enzyme.

Drugs Acting Directly on Cells

- Pilocarpine hydrochloride or pilocarpine nitrate
- Pilocarpine is the main alkaloid obtained from the leaflets of *Pilocarpus jaborandi* and *P. micropyllus*, family Rutaceae.
- Pilocarpine stimulates all smooth muscles and glands innervated by cholinergic fibers and shares many of the properties of ACh and muscarine

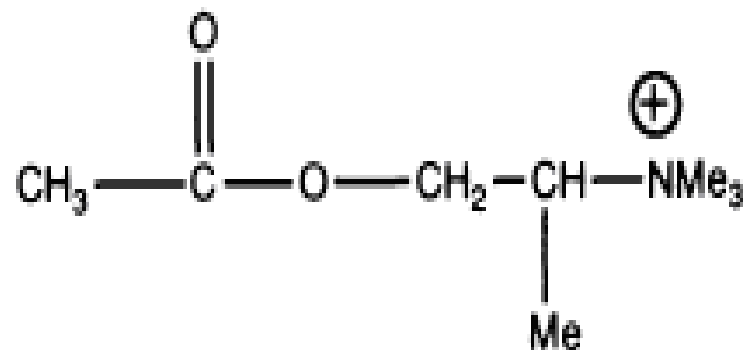
Clinical uses of cholinergic agonists

Muscarinic agonists:

- Treatment of glaucoma.
- 'Switching on' the GIT and urinary tract after surgery.
- Treatment of certain heart defects by decreasing heart muscle activity and heart rate.

Nicotinic agonists:

- Treatment of myasthenia gravis.
- An example of a selective nicotinic agonist is shown in Fig. below
 - α -methyl group

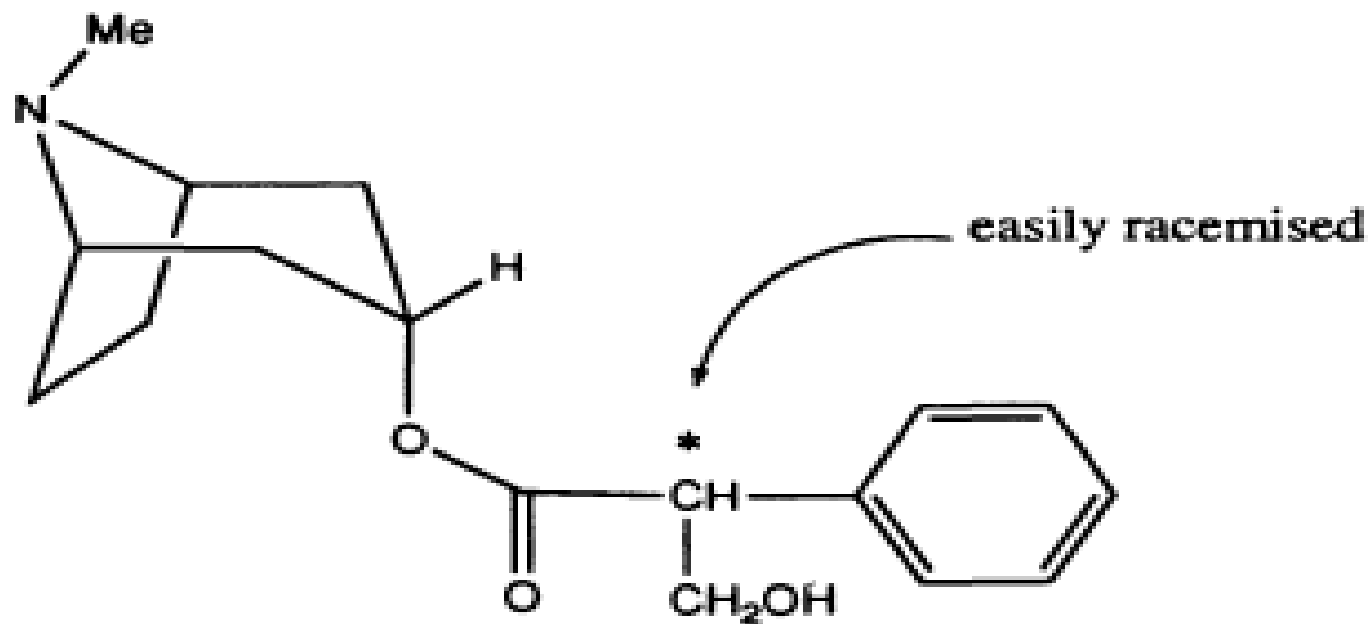


Muscarinic antagonists

- The first antagonists were natural products and in particular alkaloids.

Atropine

- Atropine has a chiral centre (*) and therefore two enantiomers are possible.
- It is present in the plant species *Solanaceae* as a single enantiomer called **hyoscyamine**.

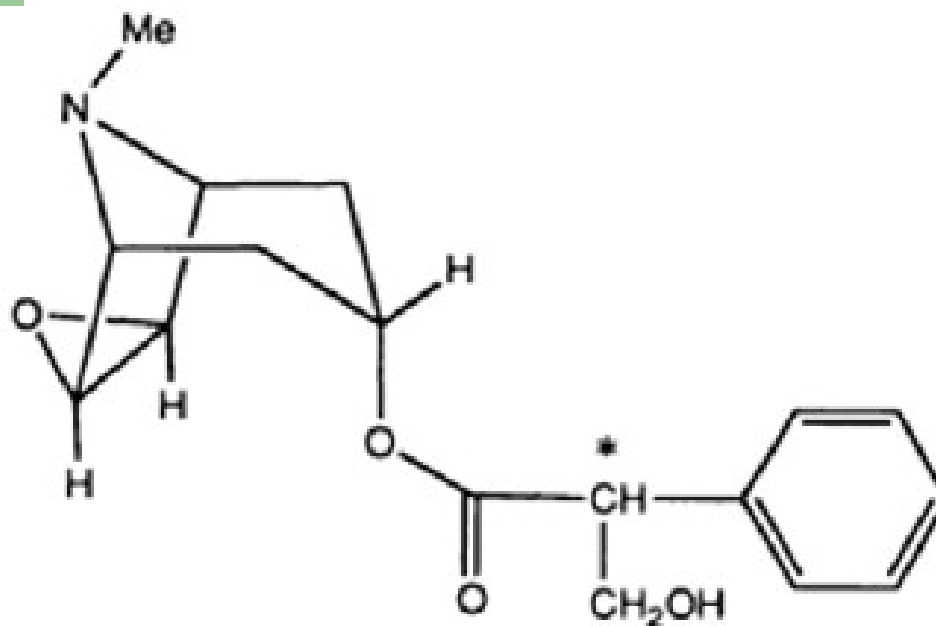


Atropine (hyoscyamine)

- However, as soon as the natural product is extracted into solution, the chiral centre racemizes such that atropine is obtained as a racemic mixture and not as a single enantiomer.
- The chiral centre in atropine is easily racemized since it is next to a carbonyl group.
 - The proton attached to the chiral centre is acidic and as a result is easily replaced.
- Atropine was obtained from the roots of belladonna (deadly nightshade) in 1831.

Hyoscine (1879-84)

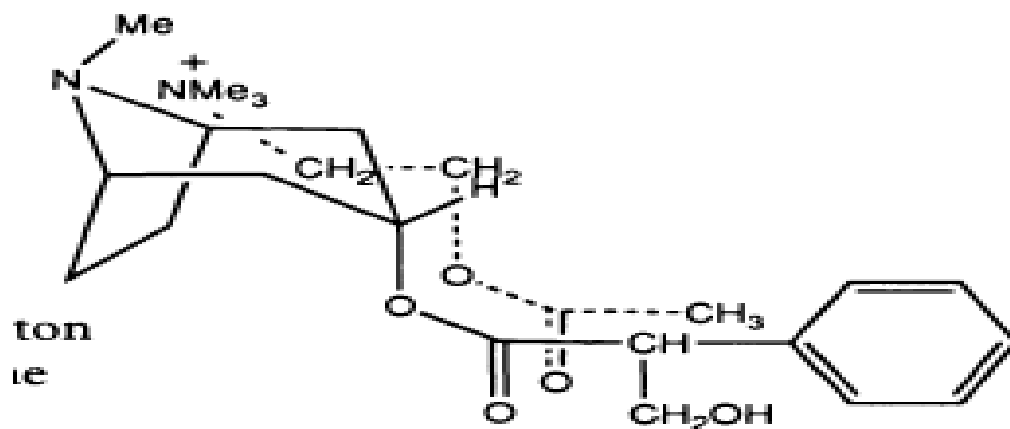
- Hyoscine (or scopolamine) is also obtained from solanaceous plants.
- In high doses, both hyoscine and atropine are **hallucinogens**



Hyoscine (scopolamine).

- These two compounds can bind to and block the cholinergic receptor.
- From their structure we can see that basic nitrogen and an ester group are present.

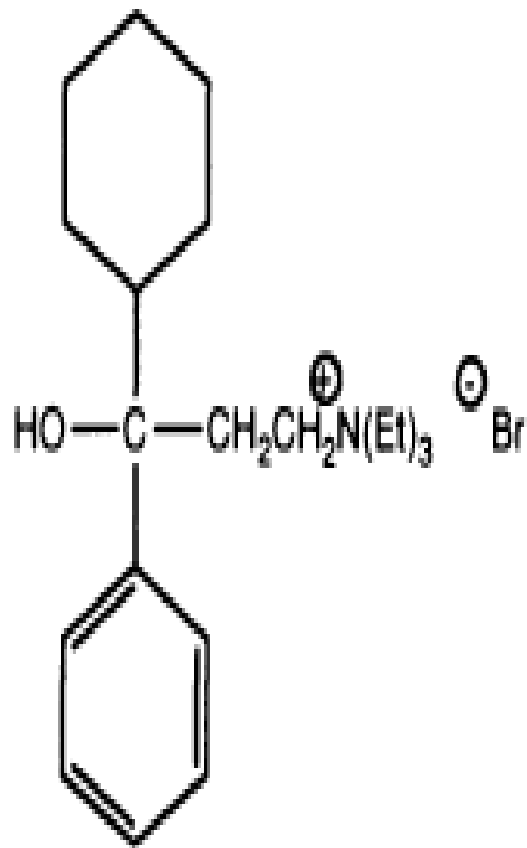
- If we superimpose the acetylcholine skeleton on to the atropine skeleton, the distance between the ester and the nitrogen groups are similar in both molecules



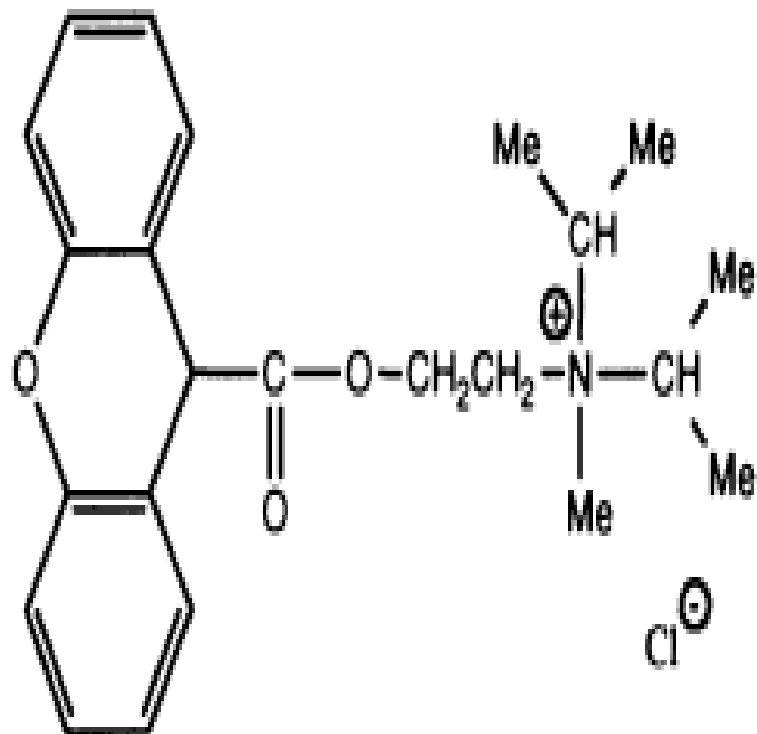
- The nitrogen atom in atropine is protonated when it binds to the cholinergic receptor.
- Atropine can be seen to have the two important binding features of acetylcholine
 - A charged nitrogen (if protonated) and
 - An ester group
- It is, therefore, able to bind to the receptor, but is unable to **'switch it on'**.

- Since atropine is a larger molecule than acetylcholine, it is capable of binding to other binding groups outside of the acetylcholine binding site.
- As a result, it interacts differently with the receptor, and does not induce the same conformational changes as acetylcholine.

- Analogues of atropine were synthesized to 'slim down' the structure to the essentials.
- This resulted in a large variety of active antagonists (e.g. tridihexethyl bromide and propantheline chloride).

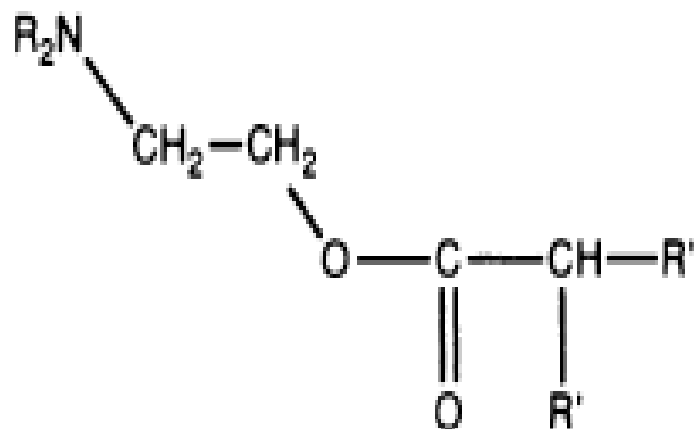


TRIDIHSEXETHYL BROMIDE



PROPANTHELINE CHLORIDE

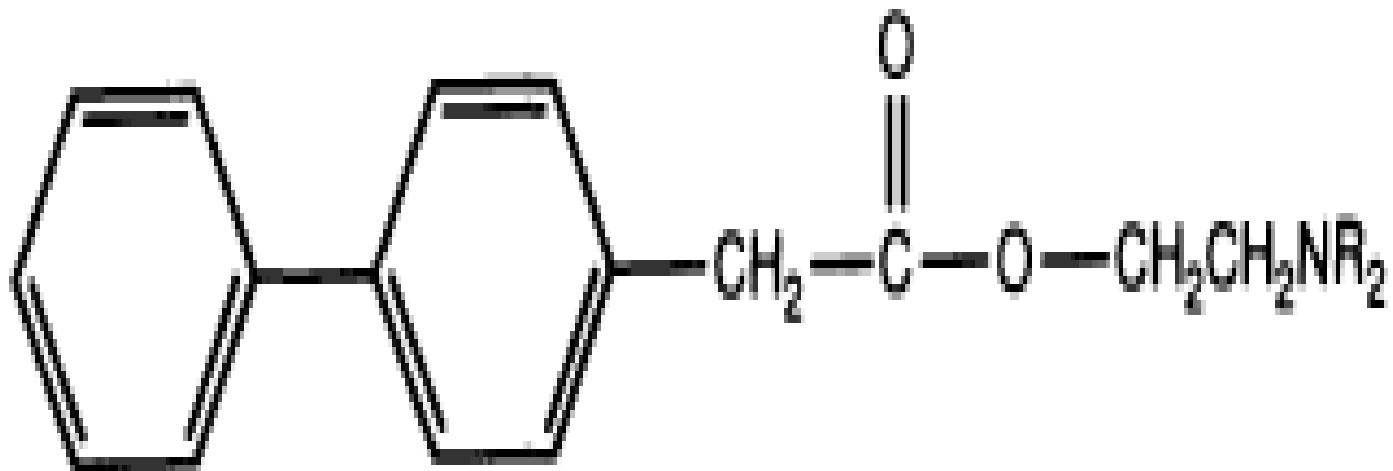
- SAR studies have come up with the following generalizations:



R' = Aromatic or
Heteroaromatic

- The alkyl groups on nitrogen (R) can be larger than methyl (in contrast to agonists).
- The nitrogen can be tertiary or quaternary, whereas agonists must have quaternary nitrogen.
- Very large acyl groups are allowed ($R' =$ aromatic or heteroaromatic rings).
 - This is in contrast with agonists where only the **acetyl** group is permitted.

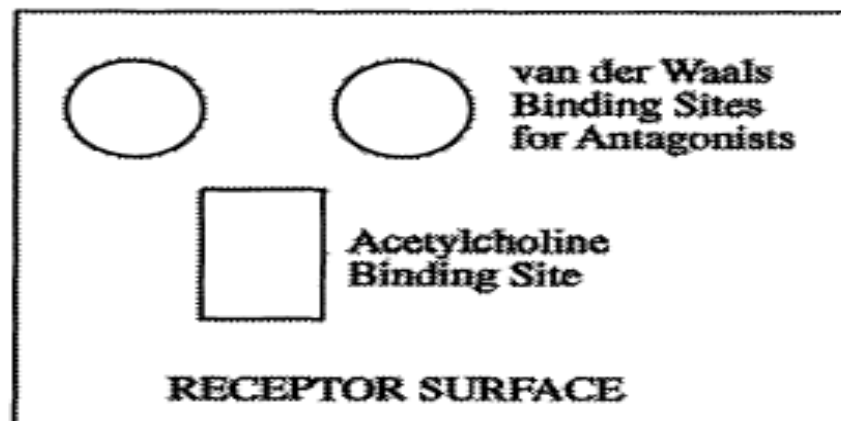
- It is the last point which appears to be the most crucial in determining whether a compound will act as an antagonist or not.
- The acyl group has to be bulky, but it also has to have that bulk arranged in a certain manner (i.e. there must be some sort of branching in the acyl group).
 - For example, the molecule shown in Fig. below has a large **unbranched acyl** group but is **not** an **antagonist**.



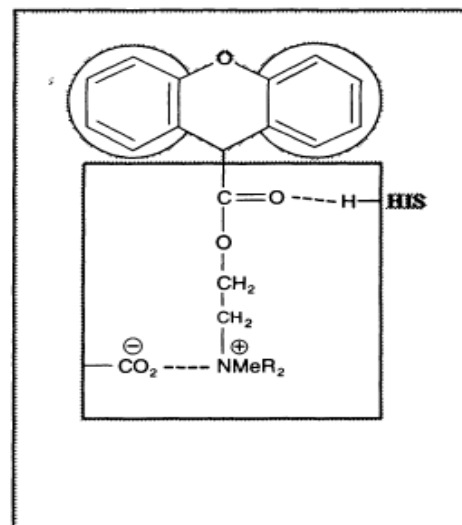
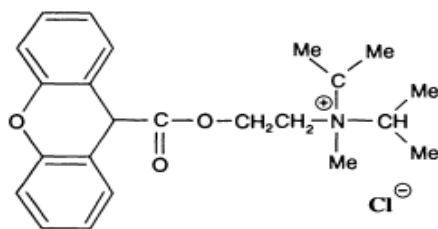
- **Conclusion:**

- There must be another binding site on the receptor surface next to the normal acetylcholine binding site.
- This area must be hydrophobic since most antagonists have aromatic rings.

- The overall shape of the acetylcholine binding site plus the extra binding site would have to be **T** or **Y-shaped** in order to explain the importance of branching in antagonists.

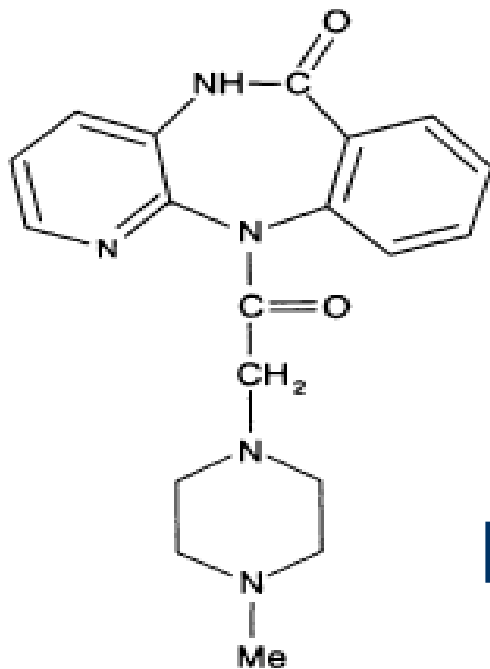


- A structure such as propantheline binds more strongly to the receptor than acetylcholine itself.



- The conformational changes induced in the receptor (if any are induced at all) will be different from those induced by acetylcholine.
- A large variety of antagonists have proved to be useful medicines, with many showing selectivity for specific organs.
- For example, some act at the intestine to relieve spasm, some act selectively to decrease gastric secretions, while others are useful in ulcer therapy.

- This selectivity of action is due to the distribution properties of the drug than to receptor selectivity i.e.
 - Compounds can reach one part of the body more easily than another
- However, the antagonist **pirenzepine**, which is used in the treatment of peptic ulcers, is a selective **M1 antagonist** with no activity against M2 receptors.



Pirenzepine

- Since antagonists bind more strongly than agonists, they are better compounds to use for the labelling and identification of receptors on tissue preparations.

- The antagonist, labelled with a radioactive isotope of H or C, binds strongly to the receptor and the radioactivity reveals where the receptor is located
- Ideally, we would want the antagonist to bind irreversibly in this situation.
 - Such binding would be possible if the antagonist could form a covalent bond to the receptor.

- One useful tactic is to take an established antagonist and to incorporate a reactive chemical centre into the molecule
- This reactive centre is usually electrophilic so that it will react with any suitably placed nucleophile close to the binding site (for example, the OH of a serine residue or the SH of a cysteine residue).

- In theory, the antagonist should bind to the receptor in the usual way and the electrophilic group will react with any nucleophilic amino acid within range.
- The resulting alkylation irreversibly binds the antagonist to the receptor through a covalent bond.

- In practice, the procedure is not always as simple as this, since the highly reactive electrophilic centre might react with another nucleophilic group before it reaches receptor binding site.
- One way to avoid this problem is to include a latent reaction centre which can only be activated once the antagonist has bound to the recap binding site.

Antagonists of the nicotinic cholinergic receptor

Applications of nicotinic antagonists

- Nicotinic receptors are present in nerve synapses at ganglia, as well as at the neuromuscular synapse.

- Antagonists of ganglionic nicotinic receptor sites are not therapeutically useful since they cannot distinguish between the ganglia of the sympathetic nervous system and the ganglia of the parasympathetic nervous system (both use nicotinic receptors).
- But antagonists of the neuromuscular junction (NMJ) are therapeutically useful and are known as neuromuscular blocking agents.

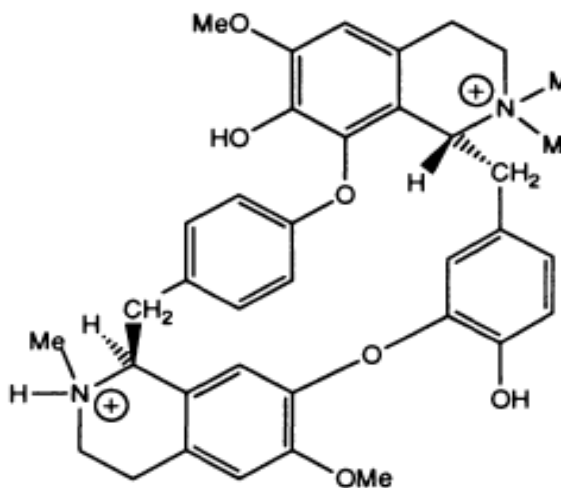
Nicotinic antagonists

- **Curare (1516) and tubocurarine**
- Curare was first identified when Spanish soldiers in South America found themselves the unwilling victims of poisoned arrows.
- It was discovered that the Indians were putting a poison on to the tips of their arrows.
- This poison was a crude, dried extract from a plant called *Chondrodendron tomentosum* and caused paralysis as well as stopping the heart.

- We now know that curare is a mixture of compounds.
- The active principle, however, is an antagonist of acetylcholine which blocks nerve transmissions from nerve to muscle.
- Can be used medically if they are taken at the right dose levels and under proper control.

- The main application is in the relaxation of abdominal muscles in preparation for surgery. (This allows the surgeon to use lower levels of general anesthetic)
- Curare, as mentioned above, is actually a mixture of compounds, and it was not until 1935 that the active principle (Tubocurarine) was isolated.

- The determination of the structure took even longer and was not established until 1970.



Important question to be raised here

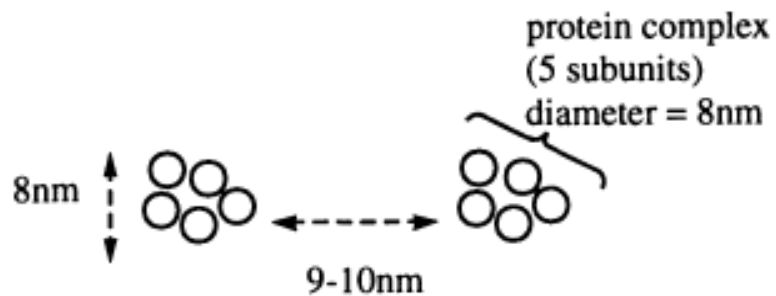
- Why should tubocurarine bind to and block the cholinergic receptor without having ester functional group?
- The answer lies in the fact that the molecule has *two* positively charged nitrogen atoms (one tertiary which is protonated, and one quaternary).

There are three assumptions

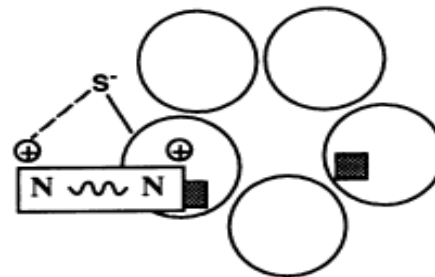
1. Tubocurarine forms a bridge between two adjacent cholinergic receptors.
 - But latter it was proved that the distance between the two receptor proteins dimer was larger than (9-10 nm) the distance between the two positively charged nitrogen centers (1.4nm).

- 
- 
2. The tubocurarine molecule bridges two acetylcholine binding sites within the one protein complex. Since there are two such sites within the complex, this appears an attractive alternative theory.
- However, the two sites are further apart than 1.4 nm and so this too seems unlikely.

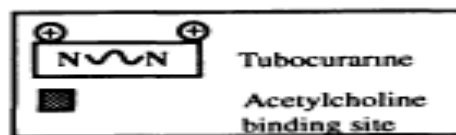
3. One of the positively charged nitrogens on tubocurarine binds to the anionic binding site of the acetylcholine receptor in the protein complex, while the other nitrogen binds to a nearby cysteine residue 0.9-1.2 nm away.



(a) Receptor Dimer



(b) Interaction with Tubocurarine



- Anyways despite the above uncertainties, the interaction is extremely strong and would more than make up for the lack of the ester binding interaction.
- It is also clear that the distance between the two positively charged nitrogen atoms is crucial to activity.
- Therefore, analogues which retain this distance should also be good antagonists.

Decamethonium and suxamethonium

- Decamethonium is a simple analogue of tubocurarine.
- It is a straight-chain molecule and as such is capable of a large number of conformations.
- The fully extended conformation would position the nitrogen centres 1.35 nm apart, which compares well with the equivalent distance in tubocurarine (1.4 nm).



- The drug binds strongly to cholinergic receptors and has proved a useful clinical agent.
- ***Disadvantages;***
 - Initially it acts as agonist- leads to a brief contraction of the muscle.

- It binds too strongly and as a result patients take a long time to recover from its effects.
- Not completely selective for the neuromuscular junction and has an effect on acetylcholine receptors at the heart. This leads to an increased heart rate and affect the blood pressure.

- So to overcome such limitations the following approaches were tried;
- Introduction of instability to the molecule
 - Incorporating timer control whereby the molecule can be switched off quickly and become inactive.

- Success was first achieved by introducing ester groups into the chain while retaining the distance between the two charged nitrogens to give **suxamethonium**.



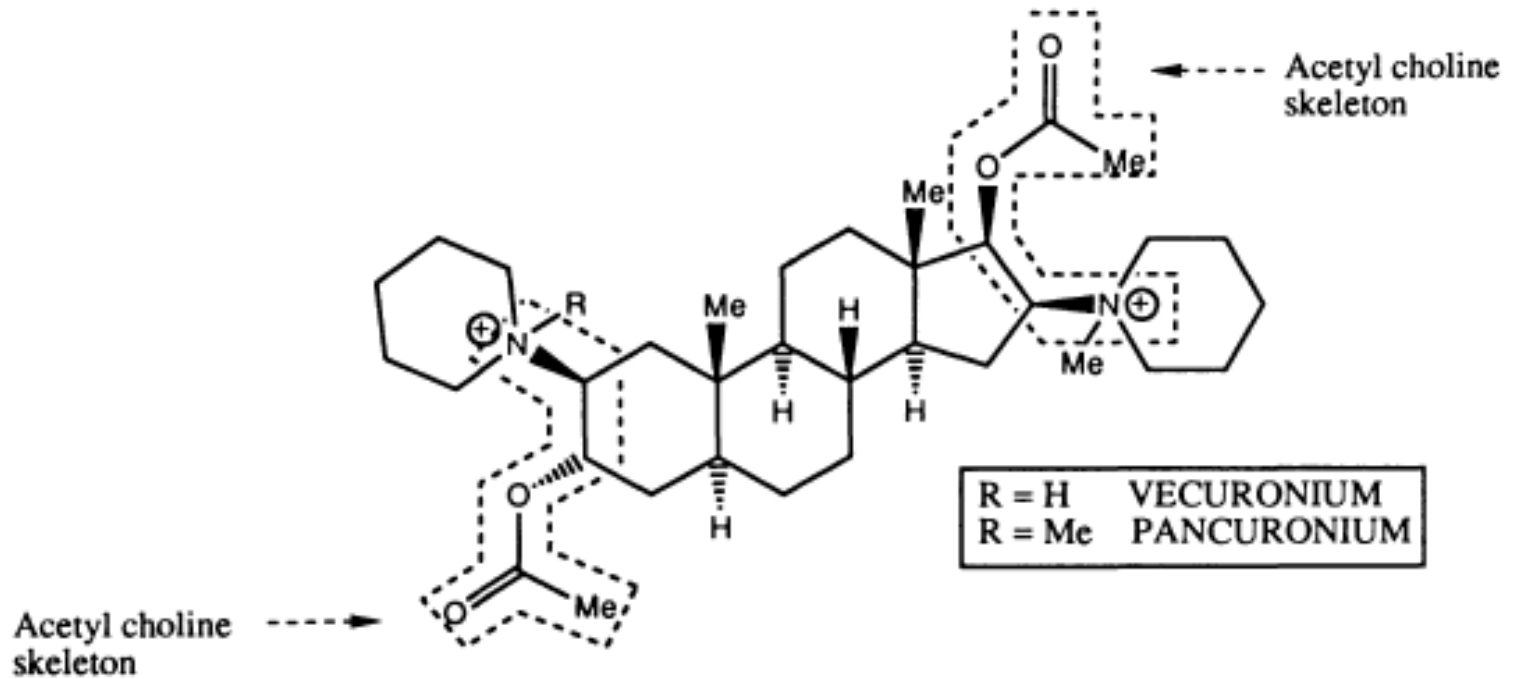
- The ester groups are susceptible to chemical and enzymatic hydrolysis.
- Suxamethonium has duration of action of five minutes, but suffers from other side-effects. (effect on autonomic ganglia)
- Furthermore, about one person in every two thousand lacks the enzyme which hydrolyses suxamethonium.

Pancuronium and vecuronium

- Pancuronium and vecuronium were designed to act like tubocurarine, but with a steroid nucleus acting as the 'spacer'.
- The distance between the quaternary nitrogens is 1.1 nm as compared to 1.4 nm in tubocurarine.

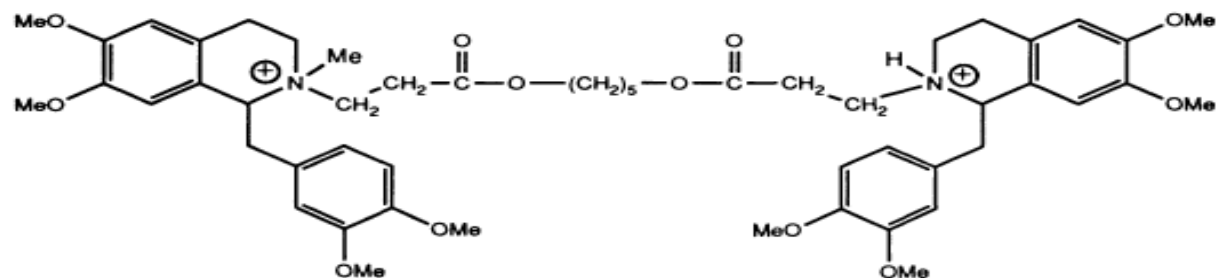
- Acyl groups were also added to introduce two acetylcholine skeletons into the molecule in order to improve affinity for the receptor sites

- These compounds have a rapid onset of action and do not affect blood pressure. However, they are not as rapid in onset as suxamethonium and also last too long (45 minutes).



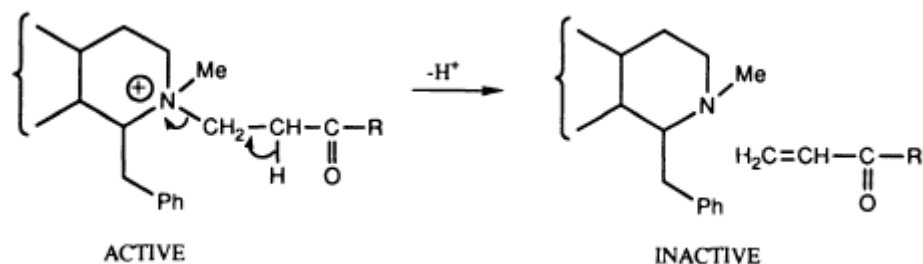
Atracurium

- The design of atracurium was based on the structures of tubocurarine and suxamethonium.
- It is superior to both since it lacks cardiac side-effects and is rapidly broken down in blood.
- This rapid breakdown allows the drug to be administered as an intravenous drip.



- The rapid breakdown was designed into the molecule by incorporating a self destruct mechanism.

- At blood pH (slightly alkaline at 7.4), the molecule can undergo a Hofmann elimination.



- Once this happens, the compound is inactivated since the positive charge on the nitrogen is lost.
- It is a particularly clever example of drug design in that the very element responsible for the molecule's biological activity promotes its deactivation.

The important features of atracurium are:

- The spacer.
 - This is the 13-atom connecting chain which connects the two quaternary centres and separates the two centres.
- The blocking units;
 - The cyclic structures at either end of the molecule block the receptor site from acetylcholine.
- The quaternary centres.
 - These are essential for receptor binding. If one is lost through Hofmann elimination, the binding interaction is too weak and the antagonist leaves the binding site.

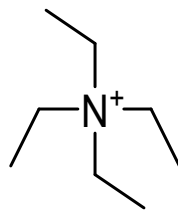
Ganglionic Blockers

- Depolarizing ganglion blockers
 - These blocking agents are actually ganglionic stimulants
 - Thus, for nicotine, small doses give an action similar to that of the natural neuroeffector ACh, an action known as the “Nicotinic effect of ACh”

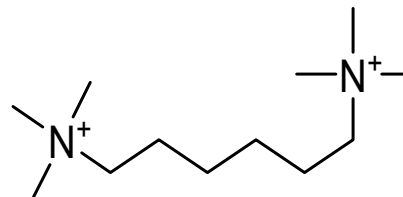
- However larger amounts of the nicotine bring about a ganglion block characterized initially by depolarization, followed by a typical competitive antagonism.
- This depolarization of the membrane of the muscle end plate is quite similar to that produced by Ach itself at ganglia and NMJ.
- Chemicals that cause this type of ganglionic block are not of therapeutic significance

- Non-depolarizing competitive ganglionic blocking agents

–Compounds in this class possess the necessary affinity to attach to the nicotinic receptor sites that are specific for Ach but lack the intrinsic activity necessary for impulse transmission.



tetraethylammonium

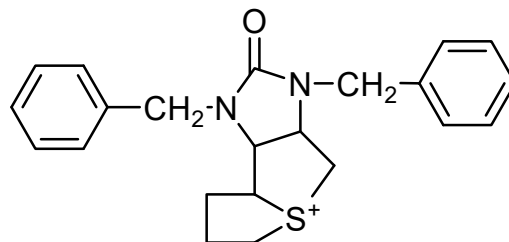


hexamethonium

- Non-depolarizing non-competitive ganglionic Blocking agents

- These blocking agents produce their effect, not at the specific Ach receptor site, but at some point further along the chain of events that is necessary for transmission of the nerve impulse.

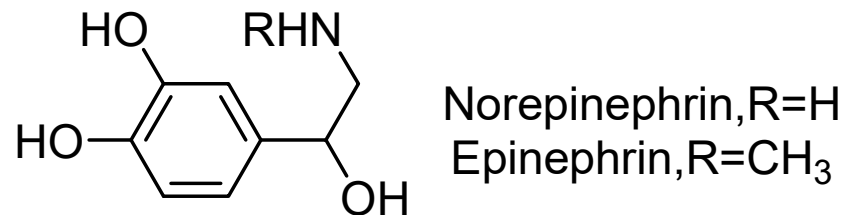
- When the block has been imposed, increase of the concentration of Ach has no effect: thus apparently, Ach does not act competitively with the blocking agent at the same receptor.



Trimethaphane camsylate

Adrenergic Neurotransmission

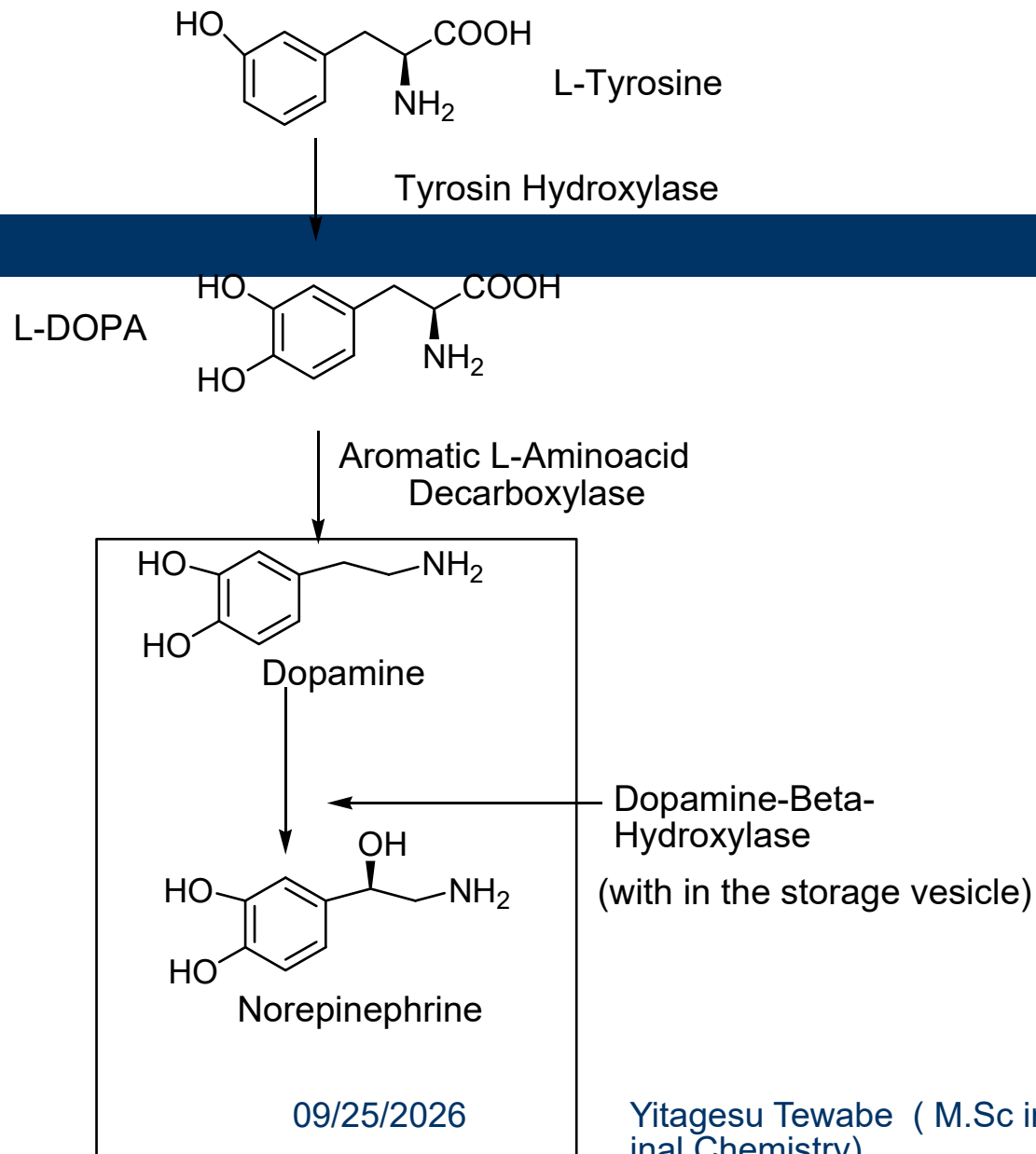
- Adrenergic drugs act on effector cells through adrenoceptors that are normally activated by norepinephrine (noradrenaline), or they may act on neurons which release the neurotransmitter.
 - Adrenergic neurons system and sympathetic neurons system used interchangeably.



- Norepinephrine and epinephrine are members of a class of pharmacologically active substances known as catecholamines.
 - Because they contain with in their structures both an amine and ortho-dihydroxy benzene (catichol).

Biosynthesis, storage and release of norepinephrine

- Biosynthesis of norepinephrine takes place within adrenergic neurons near the terminus of the axon near the junction with the effector cell.
- The biosynthesis begins with the active transport of the amino acid L-tyrosine into the adrenergic neuron

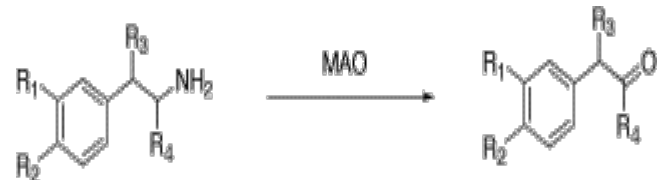
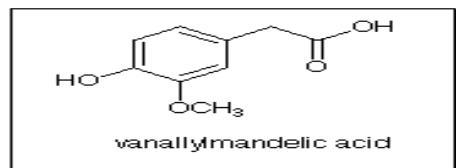
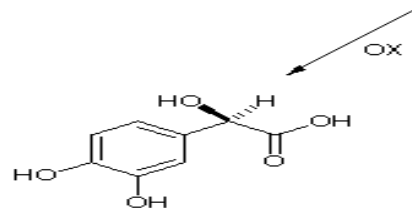
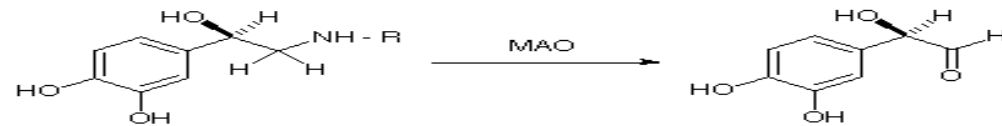


Reuptake and metabolism of norepinephrine following release

- Following its release, norepinephrine diffuses through the intracellular space to bind reversibly to adrenoceptors (α, β) on the effector cells, triggering a biochemical cascade that results in physiologic response in effector cell.

- Once it has been released and is stimulating its various receptors, there must be mechanism for removing the norepinephrin from the synapse and terminating the adrenergic impulse.
 - Recycling through active transport up take in to presynaptic neuron (uptake-1) up to 95 % of NE
 - Less efficient processes (up take -2) operates at only high conc of NE.

- That portion of released NE which escapes uptake -1 diffuses out of the synapse and is metabolized in extraneous site by catichol-O-methyl transferase, COMT, which methylates the *meta* hydroxyl group.
- Norepinephrine is also metabolized to DOPAGAL by MAO
- This pathway is also important to drug which are catechol and subject to metabolism by COMT and drugs with unsubstituted aliphatic amino group are often substrates for MAO.

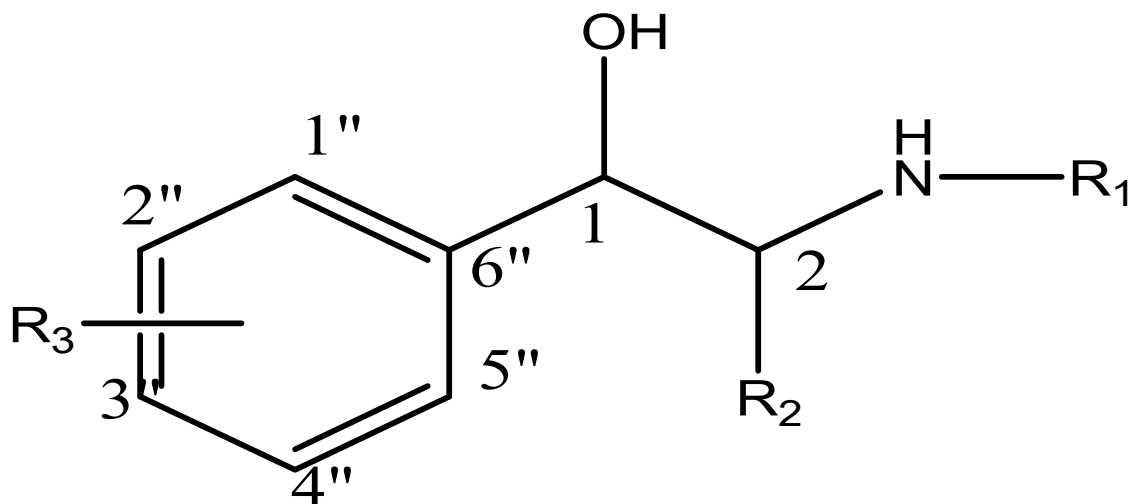


Adrenergic receptors

- α and β receptors subtypes are sufficiently well differentiated by their small molecule binding characteristics.
 - α_1 and α_2
 - β_1 , β_2 and β_3

SAR of Adrenergic agonists

- Phenyl ethanolamine agonists

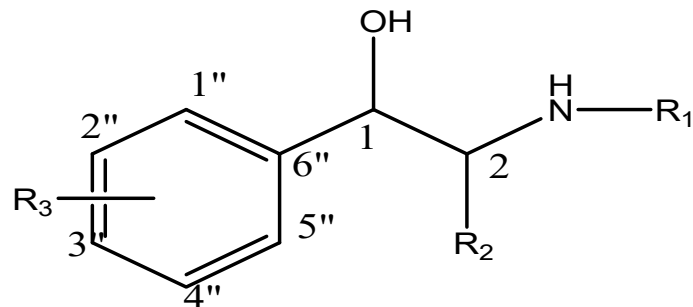


- Because of the basic amino groups, p^{ka} , range approximately 8.5-10 all of these agents are highly positively charged at physiologic p^H .
- Agents in this class have a hydroxyl group on C-1 of the side chain, β -to the amine.

- This hydroxyl substituted carbon must be in the R absolute configuration for maximal direct activity as in the natural neurotransmitter, although most drugs are currently sold as mixtures of both (R) and (S) stereoisomers at this position (racemites).
 - The nature of the other substitutes determines receptor selectivity and duration of action.

R₁ substitution on the Amino Nitrogen

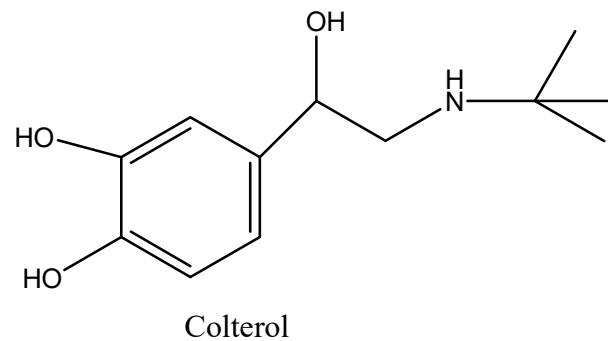
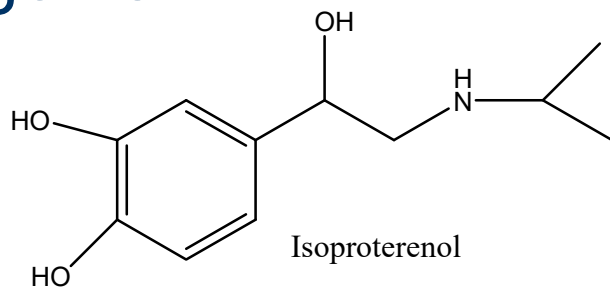
- Increase in size of R₁ from hydrogen in norepinephrine to methyl in epinephrine to isopropyl in isoproterenol,
 - Activity at α -receptors decrease and
 - Activity at β -receptors increases



- The activity at α and β -receptors is maximal when R_1 is methyl as in epinephrine, but α -agonist activity is dramatically decreased when R_1 is larger than the methyl and is negligible when R_1 is isopropyl, as in isoproterenol leaving only β -activity.

- Presumably, the β -receptor has a large **lipophilic-binding pocket** adjacent to the amine binding aspartic acid (COOH) residue, which is absent in the α receptor.
- As R_1 becomes larger than butyl, affinity for α_1 -receptor returns but not intrinsic activity, which means large lipophilic groups can afford compounds with α_1 -blocking activity e.g. Labetolol

- N- substituent can also provide selectivity for β -receptors.
 - E.g. T-butyl group $\Rightarrow$ selectivity for β_2 –receptors
- Colterol ($R_1 = t\text{-butyl}$) is a selective β_2 -agonist, whereas isoproterenol is a non-selective β -agonist.



09/25/2026

Yitagesu Tewabe (M.Sc in Medic
inal Chemistry)

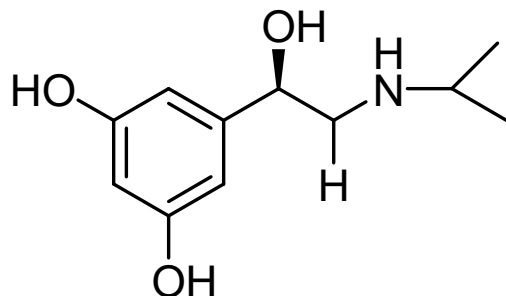
R₂-Substitution α - to basic Nitrogen, Carbon-2

- Substitution like methyl or ethyl group slow metabolism by monoamine oxidase (MAO) but has little overall effect on duration of action in catecholamines because they remain substrates for COMT.
- An ethyl group in this position diminishes α -activity far more than β -activity, affording compounds with β -selectivity such as ethylnorepinephrine.

R₃-substitution on the aromatic ring

- The natural 3',4'-dihydroxy substituted benzene ring present in norepinephrine provides excellent receptor activity for both α and β -sites.
- Both such catechol-containing compounds have poor oral activity because they are hydrophilic and rapidly metabolized by COMT.
- Alternative substitution have been found that retain good activity but are more resistant to COMT metabolism.

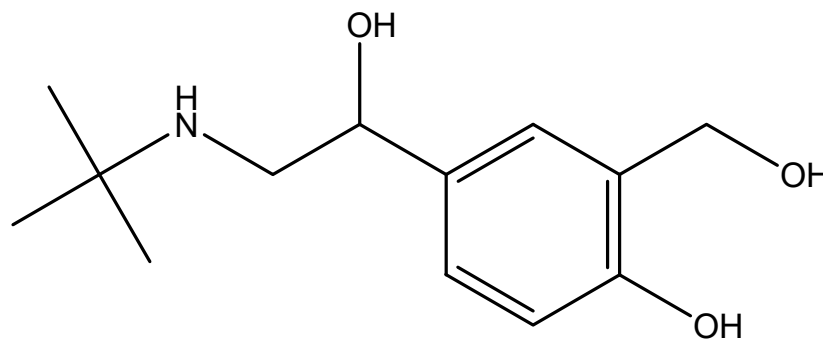
- 3',5'-dihydroxy compounds are not good substrates for COMT and in addition show **selectivity for β_2 -receptors**.



Metoproterenol

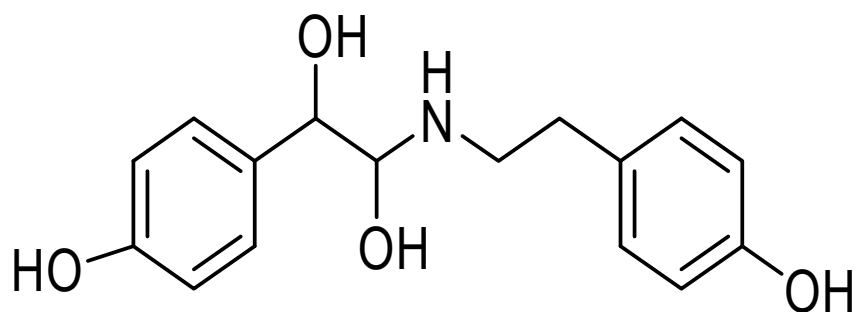
- **Metoproterenol** is orally active **bronchodilator** that have little of the cardiac stimulatory properties possessed by isoproterenol.

- Other substitution are possible that enhance oral activity and provide selectivity, such as the 3'-hydroxymethyl, 4'-hydroxyl substitution pattern of albuterol, which is also resistant to COMT.



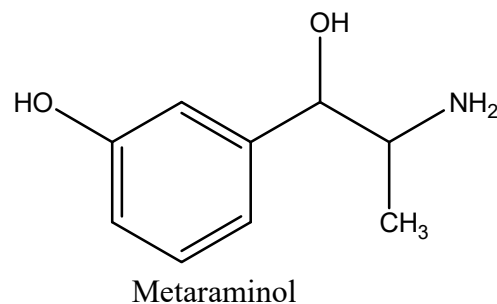
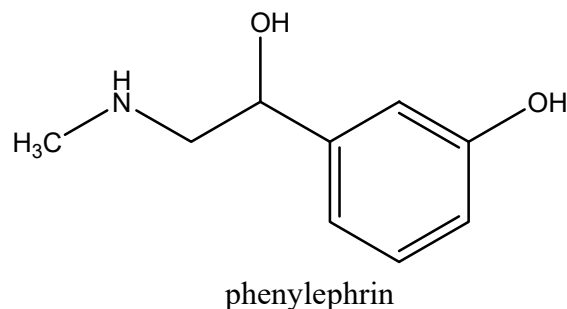
albuterol

- At least one of the groups must be capable of forming hydrogen bonds and if there is only one; it should be at the 4'-position to retain β -activity.



Ritodrine

- If R_3 is only a 3'-OH, however, activity is reduced at α sites and almost eliminated at β sites, thus selective α -agonists such as phenylephrine and metaraminol.

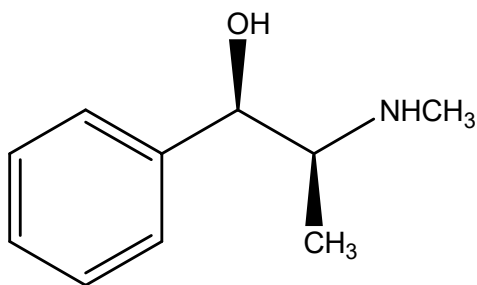


- When the phenyl ring has no phenolic substituents, ($R_3=H$) these phenyl ethanolamines may have both direct and indirect activity.

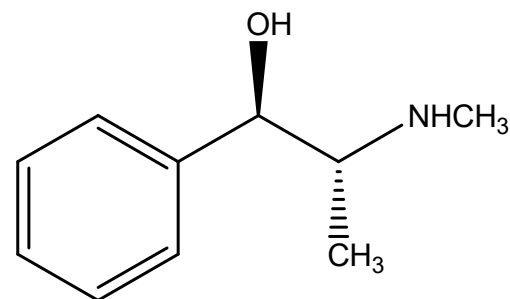
- Direct activity (i.e. agonist) is the stimulation of adrenoreceptor by the drug itself while

- Indirect activity is the result of the displacement of NE from its storage granules or receptor site inhibition resulting in non-selective stimulation of the adrenoreceptors by the displaced NE

- Since norepinephrine stimulates both α and β_1 receptors, indirect activity cannot be selective.



EPHEDRIN



PSEUDOEPHEDRIN

Norepinephrine and adrenergic drugs

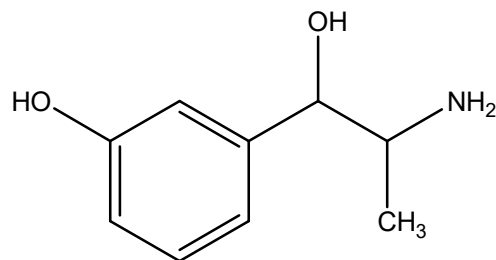
- NE has limited clinical application b/c
 - It is non-selective
 - Given only iv, because of metabolism by intestinal and liver COMT and MAO.
 - Low lipophilicity
 - Rapid metabolism limits its duration of action only 1 or 2 minutes given by infusion.

- Similar to norepinephrine, epinephrine is used to treat hypotensive crisis and because of its greater β -activity, it is used to stimulate the heart in cardiac arrest.
- Its activity on β_2
 - Administered iv and in inhalers to relieve bronchoconstriction in asthma and to application inhibiting uterine contractions.

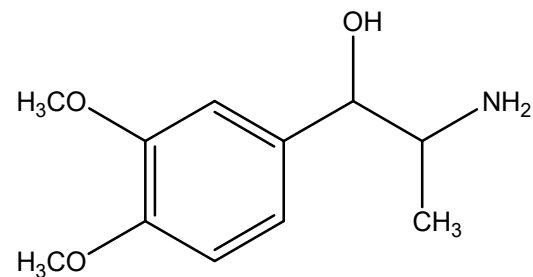
α -Adrenergic agonists

- α_1 -Agonists
- Phenyl ethanol amines
 - Meteraminol, Methoxamine and phenylephrine
- Metaraminol and methoxamine are selective α_1 receptors and have little cardiac stimulatory properties.

- Because they are not substrates for COMT, their duration of action is significantly longer than norepinephrine.



Metaraminol

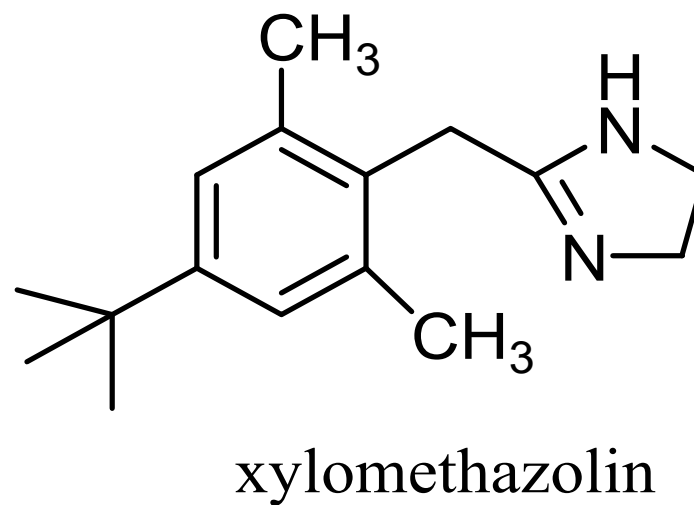
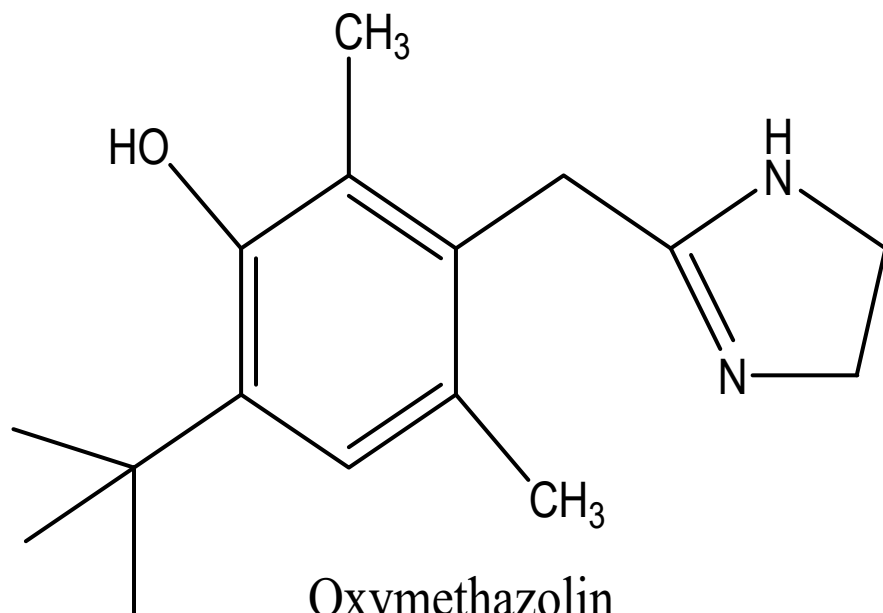


Metthoxamin

2-Arylimidazoline α_1 -agonists

- Selective α_1 -agonists and therefore they are vasoconstrictors/ vasopressors.
- They all contain a one-carbon bridge between C₂ of imidazole ring and a phenyl substituent, and therefore
 - The general skeleton of phenyl ethylamine is contained within the structures.

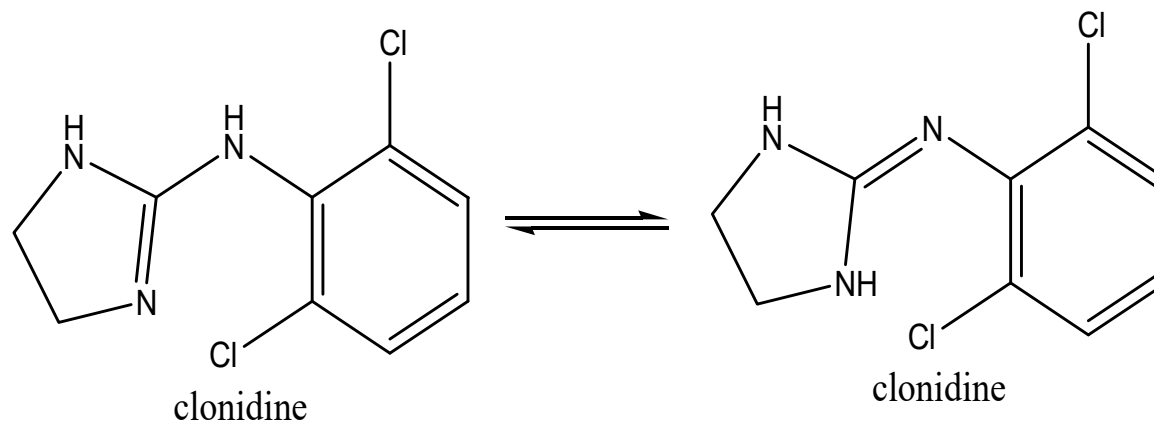
- Lipophilic substitution on the phenyl ring ortho to the methyl bridge appears to be required for agonist at α_1 and α_2 receptors.
- The bulky lipophilic groups attached to the phenyl ring at the meta or para-positions
 - Provide selectivity for the α_1 -receptor by diminishing affinity for α_2 -receptors.



α_2 -Adrenergic Agonists

- **2-aminoimidazoline**

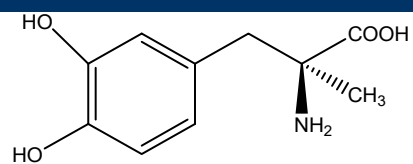
- Closely related structurally to the imidazoline structure



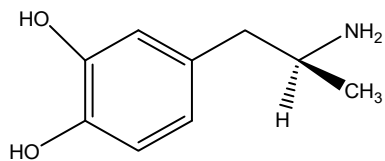
- Because of its peripheral activity on extra neuronal vascular postsynaptic α_1 -receptor,
 - Initial doses of clonidine may produce a transient vasoconstriction and an increase in blood pressure that is soon overcome by vasodilatation
- Clonidine penetrates the BBB and interact with central nervous system α_2 -receptors.

- **Methyldopa**
- Although it is structurally unrelated to the clonidine, it is antihypertensive.
- It is also a pro-drug and it is active metabolite.
- α -Methyl norepinephrine

- Methyldopa is decarboxylated by aromatic L-amino acid decarboxylase to α -methyl dopamine, which is stereo-specifically hydroxylated to 1R, 2S - α -methyl norepinephrine.

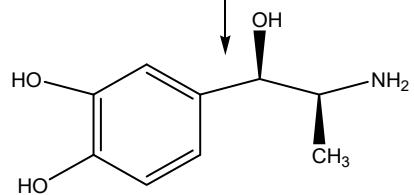


Aromatic L-aminodecarboxylase



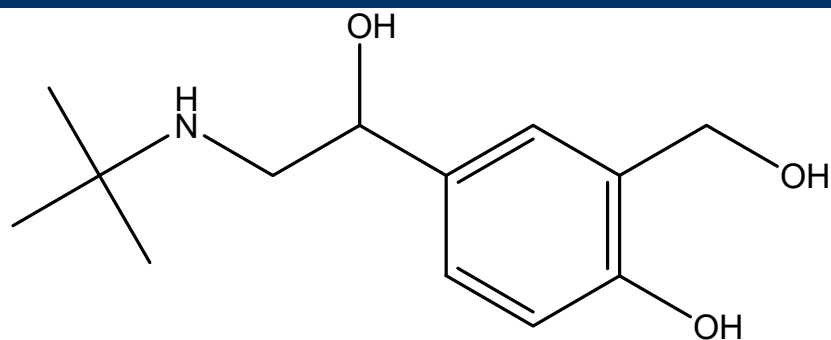
α methyltyrosine

Dopamine-beta-hydroxylase

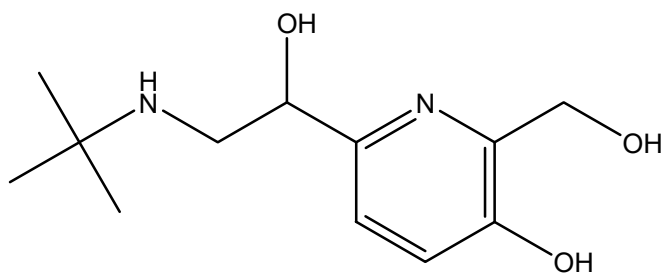


β -Adrenergic agonists

- β_2 -selective agonists
- Phenyl ethanolamines:
 - Most of β selective adrenergic agonists are used primarily as bronchodilators in asthma and other constrictive pulmonary conditions.
Albuterol, Pirbuterol, Terbutaline
 - Non-cathicol selective β_2 -agonists.
 - Available in oral dosage forms.



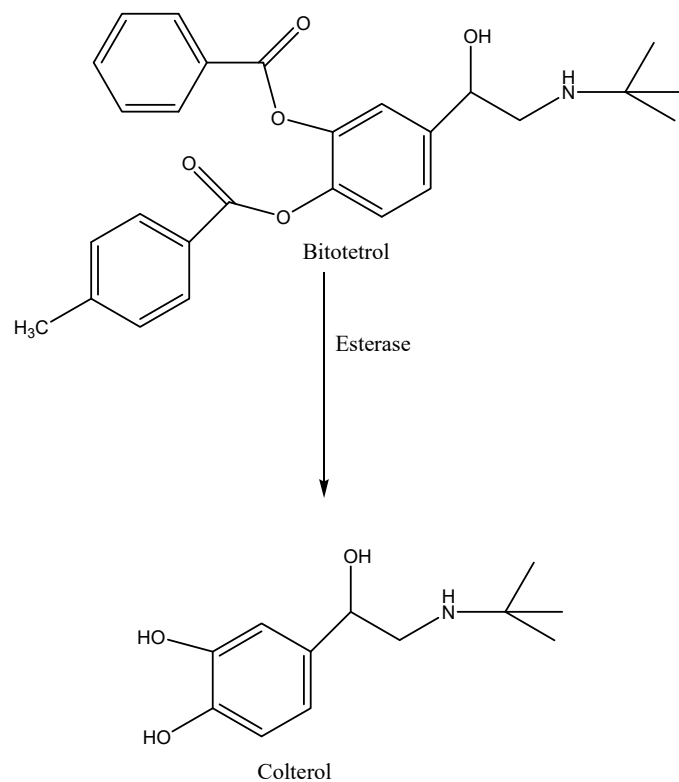
albuterol



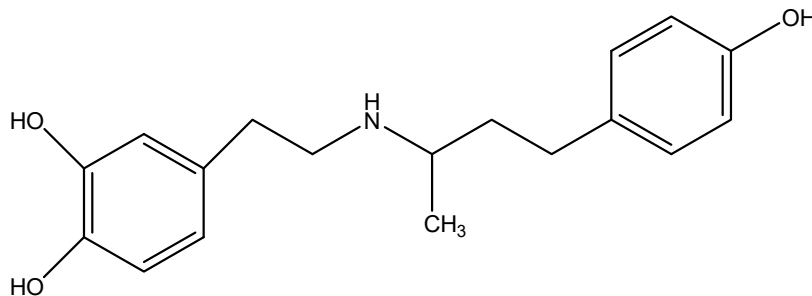
Pirbuterol

Bitolterol_

- Is a prodrug form of colterol
- Has high lipophilicity and prolonged duration of action
- The ester group is hydrolyzed by esterase to liberate the active drug, colterol.



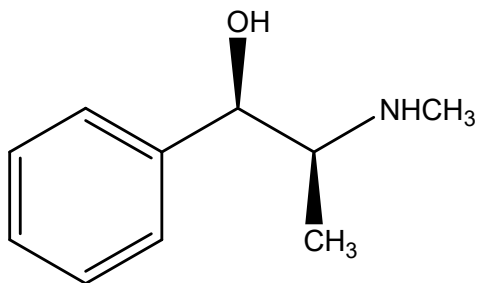
- β_1 -Adrenergic agonists:
- Dobutamine
 - Has no aliphatic β -hydroxy group.
 - Dopamine analogue with bulky arylalkyl group on the nitrogen and one chiral (assymmetric) center.



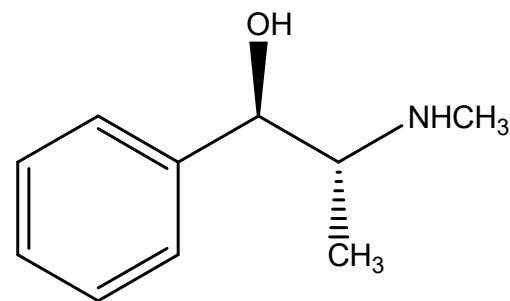
Mixed Acting Sympathomimethics

- Ephedrine
 - Ephedrine is a natural product isolated from several species of ephedra plants
 - Ephedrine does not have any phenolic substituents on the phenyl ring, giving it a mixed acting response and good oral activity because it is not substrate for COMT

- Lacking hydrogen bonding phenolic substituents, ephedrine is less polar and can easily pass BBB



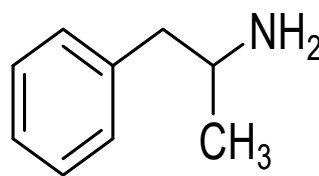
Ephedrine



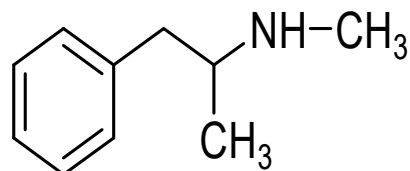
Pseudoephedrine

- Amphetamine and Methamphetamine
 - Methylsubstituted phenylethylamine (phenylisopropylamine), such as S(+) amphetamine and S(+)-Methamphetamine which lack both ring substituents and side chain hydroxyl group.

- Sufficiently lipophilic to cross the BBB & cause dramatic central nervous system stimulation, which gives them serious abuse potential



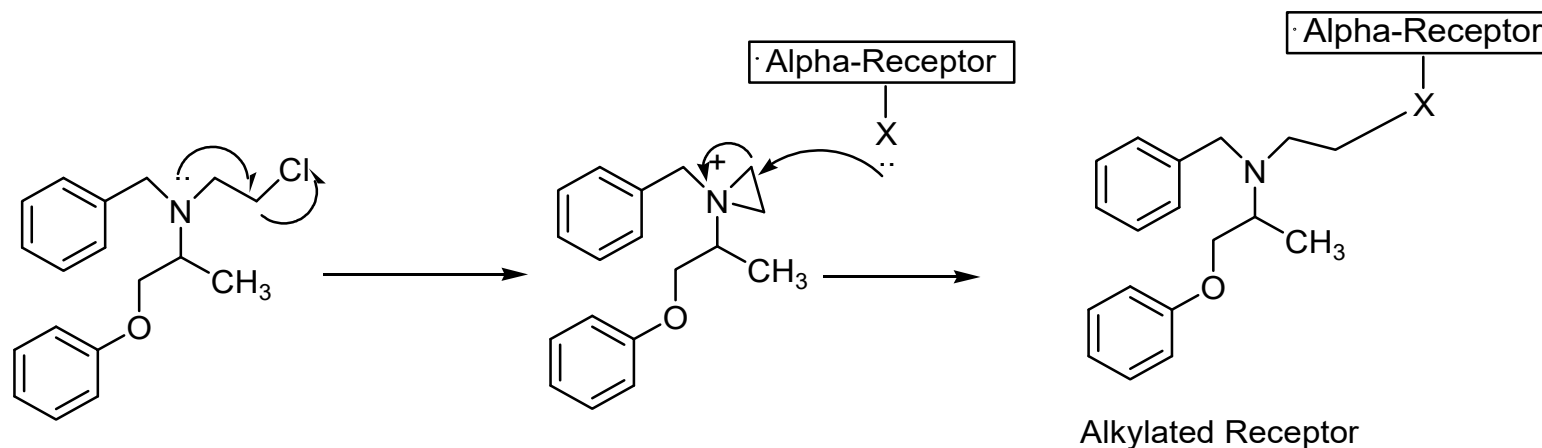
Amphetamin



Methamphetamin

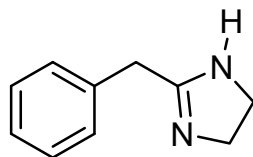
SAR of Adrenergic Antagonists

- General α -antagonists
- Phenoxybenzamine
 - α -agonists cause vasoconstriction and raise blood pressure
 - α -antagonists to be therapeutically used as antihypertensive agents
- phenylbenzamine(dibenzylamine)
 - a β -haloalkylamine that alkylate α -receptors.

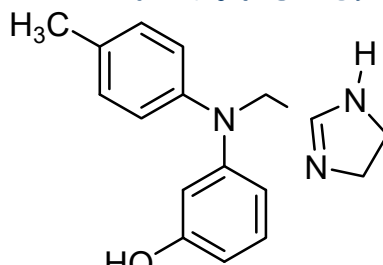


- Phenoxybenzamine alkylation of alpha receptors, X- is Nucleophile such as S,N,O (Part of amino acid)
- Unfortunately, other biomolecules besides the target α -receptors are also alkylated

• Tolazoline and Phentolamine



Tolazoline

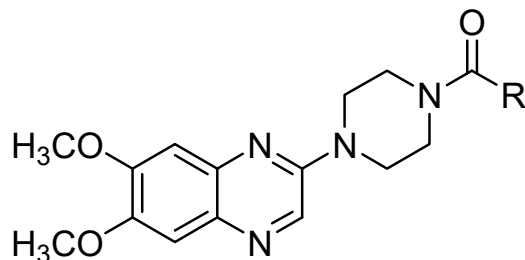


Phentolamine

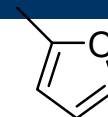
- Are two imidazoline α -antagonists that also have antihypertensive activity, although they have been replaced by far better agents

Selective α_1 -Antagonists

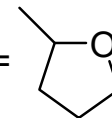
- Quinazolines:



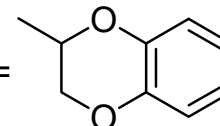
Prazosin:R =



Terazosin:R =



Doxazosin:R =

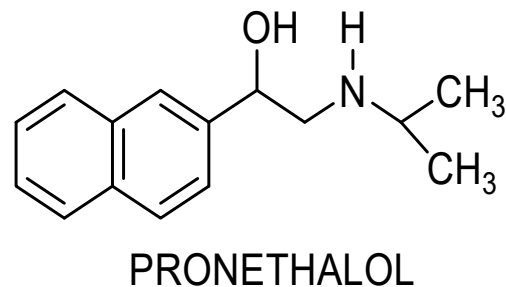
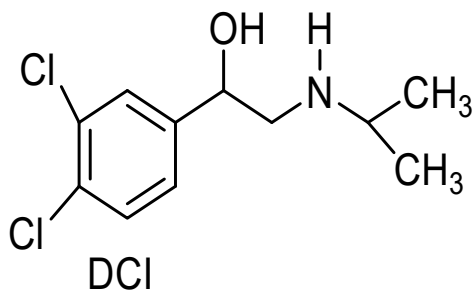


- Prazocin, doxazosin and terazosin contain 4-amino-6,7-dimethoxyquinazoline ring system attached to piperazine nitrogen
- The only structural difference is to the structure attached to the other nitrogen of the piperazine
 - The difference in these groups affords dramatic differences in pharmacokinetic properties of these drugs

- When the furan ring of prazosin is reduced to form tetrahydrofuran ring of terazocin, the compound becomes significantly hydrophilic.

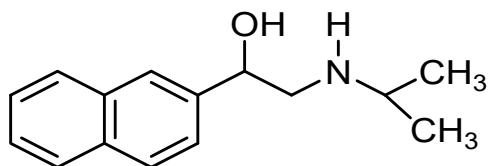
β -Adrenergic antagonists

- A derivative of isoproterenol in which the catichol hydroxyls had been replaced by chlorines, dichloroisoproterenol (DCI) was discovered to be β -antagonist
- DCI- had no clinical utility, replacement of the 3,4-dichlorosubstituents with a carbon bridge to form a naphthalylethanolamine derivative gave a clinical candidate, Pronethalol

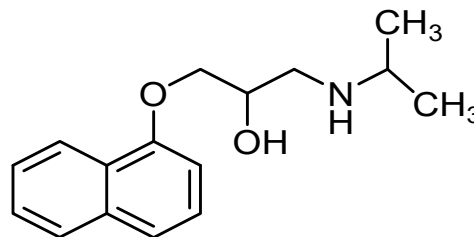


Structure Activity Relationship

- Introduction of oxy methylene bridge OCH_2 , into the aryl ethanolamine structure of pronethalol gives propranolol



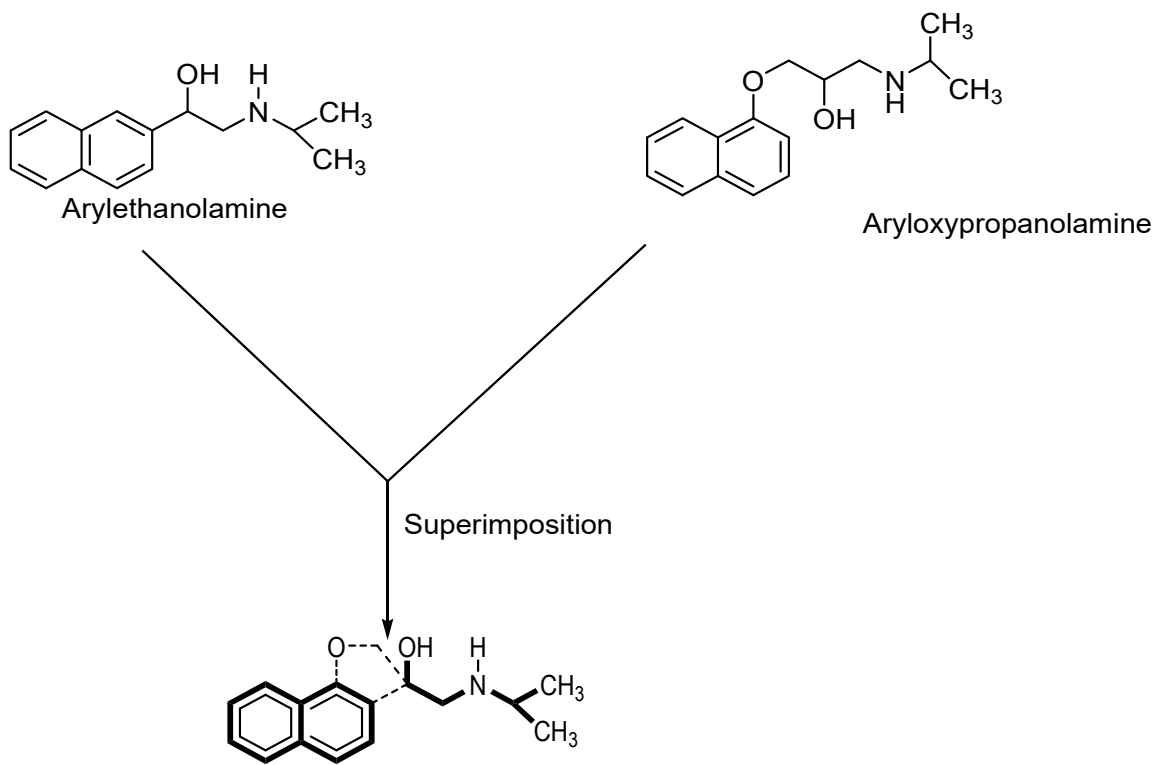
Pronethalol



Propranolol

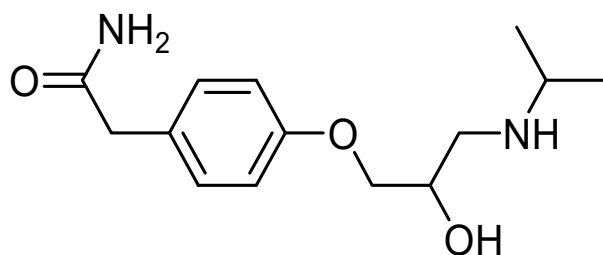
- In general, the aryloxy propanolamines are more potent β -blockers than the corresponding aryl ethanolamines and most of the β -blockers currently used clinically are aryloxypropanolamines

- Initially it might appear that lengthening the side chain would prevent appropriate binding of the required functional groups to the same receptor.
 - But molecular modeling show that the side chains of aryloxypropanolamines can adopt a conformation that places the hydroxyl and amine groups into approximately the same position.

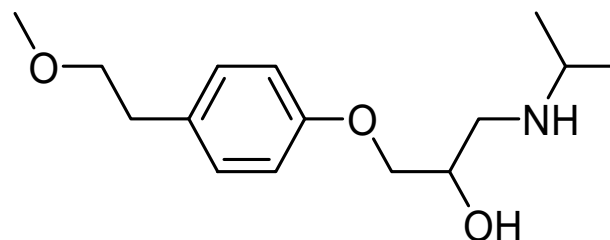


Propranolol

- Was originally discovered for the treatment of angina pectoris
- During clinical trial as antianginal, propranolol was discovered to have antihypertensive property
- A new series of 4-substituted phenyloxy- propanolamines emerged such as practolol, which selectively inhibited sympathetic cardiac stimulation(β_1 -selective).



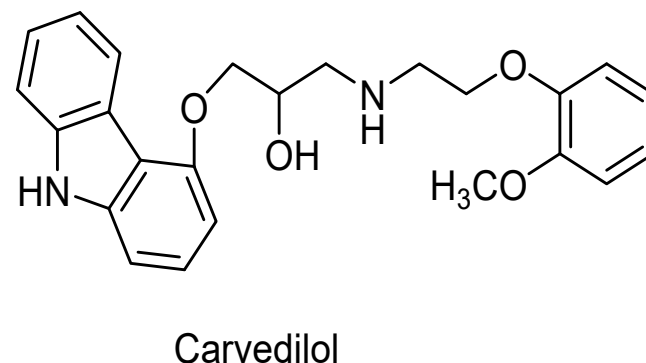
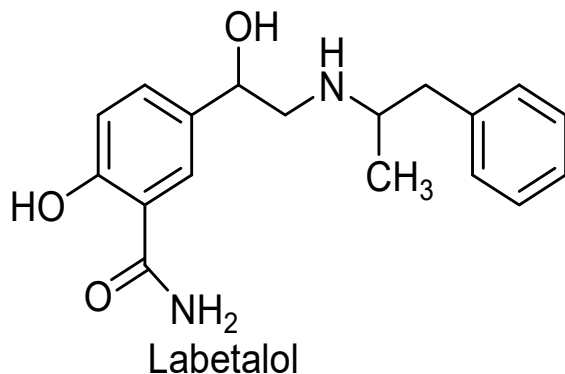
Atenolol



metoprolol

Mixed α/β Adrenergic Antagonists

- Labetalol and Carvedilol
- They have unusual activity in that they are antihypertensive with $\alpha_1, \beta_1, \beta_2$ blocking activity



- Groups such as isopropyl and t-butyl eliminated α -receptor activity, still larger groups could bring back α_1 -affinity but not intrinsic activity.
- Thus these two drugs have structural features permitting binding to both the α_1 and non-selectively to both β -receptors.



End